



# 1,3-Propanediol extraction from aqueous phase using hydrophobic eutectic solvents: Insights from experiments and molecular dynamics simulations

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## ABSTRACT

1,3-Propanediol (1,3-PDO) is a pivotal chemical with diverse industrial applications in polymers, cosmetics, and pharmaceuticals. However, current challenges in its downstream processing demand innovative solutions with an emphasis on sustainability. This study explores using hydrophobic eutectic solvents to address the critical separation challenges of 1,3-PDO from aqueous systems more efficiently and sustainably. Five different eutectic solvents consisting of DL-menthol as hydrogen bond acceptor (HBA) and dodecanoic acid, thymol, and palmitic acid as hydrogen bond donors (HBDs), one eutectic solvent comprising of thymol as HBA and dodecanoic acid as HBD were synthesized and studied for 1,3-PDO extraction from the aqueous solution. All the studied solvents showed excellent results, with distribution coefficient ( $\beta$ ) ranging from 20 to 234 and selectivity ( $S$ ) ranging from 480 to 18000, with DL-menthol: dodecanoic acid (3:1) being the most efficient solvent. Radial distribution function (RDF), spatial distribution function (SDF), and hydrogen bonding analyses performed via molecular dynamics simulation corroborated the experimental findings and highlighted the favorable interactions of the eutectic solvents with 1,3-PDO due to the formation of hydrogen bonds between them. The lower activity coefficient predicted by the COSMO-SAC model indicates strong solute-solvent interaction, facilitating efficient 1,3-PDO extraction. In contrast, higher activity coefficient values for water indicate weaker interactions, emphasizing the effectiveness of eutectic solvents in extracting 1,3-PDO from water.

## 1. Introduction

1,3-Propanediol (1,3-PDO) is a versatile C3 platform chemical obtained from glycerol hydrogenolysis. It is a key monomer in the polymer industry for producing polyesters, polyethers, and polyurethanes. Additionally, 1,3-PDO is essential in creating poly(trimethylene terephthalate) and polypropylene terephthalate, both of which are biodegradable polyesters used in textiles and carpets. Beyond its polymer applications, 1,3-PDO is utilized in cosmetics, personal care products, cleaning agents, lubricants, and pharmaceuticals [1]. It also plays a role in producing heterocyclic compounds. It is used in engine coolants, food and beverage processing, de-icing fluids, water-based inks, heat transfer fluids, and unsaturated polyester resins [2]. The global 1,3-PDO market was valued at USD 491.97 million in 2023 and is projected to grow at a compound annual growth rate of 7.2 % from 2023 to 2032 [3]. Traditionally, the production of 1,3-PDO has relied on chemical synthesis methods developed by companies such as Shell and Degussa (now owned by DuPont) [4]. The Shell method utilizes ethylene oxide in a

two-step process involving hydroformylation to produce 3-hydroxypropionaldehyde, followed by hydrogenation to yield 1,3-PDO. Alternatively, the Degussa-DuPont method uses acrolein as the starting material, which is hydrated to form 3-HPA and then hydrogenated to produce 1,3-PDO [5]. These chemical synthesis routes are petroleum-based and require high pressure, high temperature, and catalysts, leading to drawbacks such as the generation of toxic intermediates and harsh processing conditions.

In response to these issues, recent efforts have shifted towards biological pathways that utilize renewable biomass and microorganisms for 1,3-PDO production. Microbial conversion of raw materials presents a more environmentally friendly approach, avoiding the production of toxic by-products and operating under milder conditions [6]. Fermentation processes provide significant flexibility in selecting feedstocks, particularly low-cost substrates like glycerol from biodiesel production. This approach not only facilitates waste valorization but also contributes to lowering production costs [7]. Additionally, metabolic engineering can be employed to optimize microbial pathways, thereby improving the

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yield and productivity of 1,3-PDO. The scalability of biotechnological processes, combined with advancements in downstream processing, positions fermentation as a viable and sustainable route for large-scale 1,3-PDO production. However, despite these advantages, significant challenges remain, especially in the downstream processing stage, which accounts for a large portion of the total production cost. Amidst these challenges, innovative approaches in downstream processing hold promise for overcoming the hurdles encountered in traditional separation methods. By refining techniques such as liquid–liquid extraction (LLE), researchers aim to enhance the efficiency and sustainability of 1,3-PDO production. For instance, in a study conducted by Boongsawat et al. (2010), using ethyl acetate as solvent and ethanol as cosolvent (9:1) yielded a distribution coefficient ( $\beta_{1,3\text{-PDO}}$ ) value of 0.20 and selectivity ( $S_{1,3\text{-PDO}}$ ) value of 1.17 [8]. In another research effort focusing on multiple ionic liquids (ILs), 1-butyl-3-methylimidazolium methylsulfate demonstrated optimal performance in aqueous two-phase extraction with  $\text{K}_2\text{HPO}_4$  and  $\text{KH}_2\text{PO}_4$ , resulting in a  $\beta_{1,3\text{-PDO}}$  value of 7 [9]. Moreover, liquid–liquid extraction employing a tri-component extractant comprising *n*-octanol, undecanol, and tetradecanol in a 15:4:6 M ratio achieved a remarkable 98.86 % recovery of 1,3-PDO after a four-stage process [10]. Additionally, the salting-out extraction system using ethanol and  $\text{K}_2\text{HPO}_4$  exhibited notable efficiency in selectively separating 1,3-PDO, achieving a selectivity ( $S$ ) of 4.870 [11]. Furthermore, phosphonium-based ILs were employed for the liquid–liquid extraction (LLE) of 1,3-PDO, with  $[\text{P}_{66614}][\text{C}_8\text{SO}_3]$  displaying a distribution coefficient ( $\beta_{1,3\text{-PDO}}$ ) of 0.219 and a selectivity ( $S_{1,3\text{-PDO}}$ ) of 2.65 [12]. Various other industrial separation methods include crystallization, distillation, and liquid–liquid extraction (LLE), with the latter being favored for its efficiency and environmental benefits, driving advances in LLE technology [13,14]. Recent research has emphasized the utilization of imidazolium-based ionic liquids (ILs) as extractants for recovering 1,3-PDO from glycerol fermentation broth. These studies have showcased the outstanding extraction performance of ILs [9,15–17]. However, the practical application of ILs is impeded by their high production costs. To overcome this obstacle, identifying cost-effective and readily preparable IL analogues is paramount for efficiently extracting 1,3-PDO. Eutectic solvents (ESs), in particular, offer an alternative to ionic liquids and provide several advantages.

Eutectic Solvents (ESs) represent a new class of solvents produced by combining a hydrogen bond donor (HBD) and a hydrogen bond acceptor (HBA). The unique characteristic of ESs lies in the specific mixture of HBD and HBA, which creates a clear and stable eutectic liquid through hydrogen bonding interactions, resulting in a lower melting point than the individual components [18–20]. The formation and properties of ESs depend on the choices of HBD and HBA and their molar ratios. HBDs and HBAs can donate and accept protons, facilitating the formation of hydrogen bonds with solutes and thereby enhancing the solvation capabilities of the ESs [18,21,22]. This unique solvation property makes ESs highly effective for separation processes [23–29]. However, many ESs developed so far are hydrophilic, making them unstable in water-based environments and limiting their practical applications. Given the expanding range of potential uses for ESs, there is growing interest in their synthesis. It has spurred recent research into synthesizing hydrophobic eutectic solvents, which are more stable in aqueous environments and thus suitable for broader applications.

Hydrophobic eutectic solvents were first introduced in 2015 by Van Osch and colleagues, who prepared them using quaternary ammonium salts combined with decanoic acid [30]. Among various types of hydrophobic eutectic solvents, those based on menthol have proven remarkably successful [31–33]. In this context, DL-menthol is often used as the HBA, while organic acids such as lauric and decanoic acids serve as HBDs. These menthol-based eutectic solvents exhibit excellent solvation properties for polar and non-polar compounds, remain stable at high temperatures, and are suitable for extracting a wide range of inorganic and organic substances. Eutectic solvents have gained attention as a potential substitute for ionic liquids (ILs) due to their ease of

preparation, cost-effectiveness, and high purity [34]. Furthermore, eutectic solvents offer customizable properties, making them an attractive option as environmentally friendly solvents across various applications. Although many eutectic solvents can be produced for numerous applications, selecting the most suitable one often involves lengthy, complex, and expensive experimental procedures. To address this, the rapid advancement of quantum chemistry (QC) calculations [35] and Molecular dynamics (MD) simulations [36] has rendered these techniques indispensable for investigating particle structure and dynamics, effectively elucidating interaction mechanisms in various chemical processes, including separation and reaction [37–39]. Verma et al. used MD simulation to assess the phase equilibria of DES-alcohols-water, finding less than a 6 % deviation from experimental data [32]. In our previous study, MD simulations were employed to study the extraction efficiency of imidazolium-based ionic liquids for extracting 1,3-PDO from an aqueous system, with  $[\text{Bmim}][\text{SCN}]$  showing the highest distribution coefficient ( $\beta$ ) value of 15.08 and selectivity ( $S$ ) of 29.00 [15].

This work used three approaches, experimental, molecular dynamics, and COSMO-SAC thermodynamic model, to study the extraction of 1,3-PDO from aqueous solution using hydrophobic eutectic solvents. Six different types of eutectic solvents, namely, DL-menthol: dodecanoic acid (1:1), DL-menthol: dodecanoic acid (2:1), DL-menthol: dodecanoic acid (3:1), DL-menthol: thymol (1:1), DL-menthol: palmitic acid (12:1) and thymol: dodecanoic acid (1.25:1) were synthesized (based on studies reported in the literature [34,40–42].) and characterized by different analytical techniques. Firstly, ES-water binary systems were investigated to assess the stability and hydrophobicity of synthesized ESs. After that, ES-1,3-PDO-water ternary systems were studied, and vital parameters such as distribution coefficient and selectivity were evaluated based on composition analysis. Molecular dynamics simulations were carried out to calculate macroscopic parameters such as composition, microscopic properties, and interactions within the components of the binary/ternary systems. These simulations analyzed radial distribution functions (RDF), spatial distribution functions (SDF), hydrogen bonding, and mean squared displacement (MSD). Furthermore, the COSMO-SAC thermodynamic model was employed to determine the infinite dilution activity coefficient of the solute in the solvent.

## 2. Materials and methods

### 2.1. Chemicals

DL  $\pm$  Menthol ( $\geq 99.0$  %) was purchased from Sigma-Aldrich, India. Thymol ( $\geq 99$  %), dodecanoic acid ( $\geq 99$  %), palmitic acid ( $\geq 99$  %), and 1,3-propanediol ( $\geq 99$  %) were supplied by Merck, India. Dimethyl sulfoxide- $d_6$  ( $\text{DMSO-}d_6$ ,  $\geq 99.8$  %) NMR grade was purchased from SYNMR Chemicals Ltd., India. All the chemicals were analytical grade and used without further purification. The purities of all the chemicals were checked by  $^1\text{H}$  NMR spectroscopy, and peak analysis indicated negligible impurities. Millipore water was used in all the experiments.

### 2.2. Eutectic solvent synthesis and characterization methodology

Six eutectic solvents were prepared following the procedures outlined in the literature [20] and summarized in Table 1. Three DL-menthol-based ESs were synthesized by mixing DL-menthol with dodecanoic acid in three different molar ratios of 1:1, 2:1, and 3:1. Additionally, two more ESs were synthesized by mixing DL-menthol with thymol and palmitic acid in molar ratios of 1:1 and 12:1, respectively. An ES based on thymol and dodecanoic acid was synthesized by combining them in a molar ratio of 1.25:1. The ESs were synthesized using the experimental setup shown in Fig. S1. The mixture was heated at  $70^\circ\text{C}$  with a continuous stirring until a homogeneous transparent liquid was obtained. Subsequently, the ESs were cooled and subjected to vacuum drying at  $60^\circ\text{C}$  for 36 h to eliminate moisture content.

**Table 1**Eutectic solvents and their physical properties, including melting point ( $T_m$ ), density ( $\rho$ ), and water content at 298.15 K.

| HBA        | HBD             | HBA/HBD molar ratio | ESs       | Water content (wt%) | Melting point ( $T_m$ ) (°C) | Density (g/cm <sup>3</sup> ) | Density* (g/cm <sup>3</sup> ) |
|------------|-----------------|---------------------|-----------|---------------------|------------------------------|------------------------------|-------------------------------|
| DL-Menthol | Dodecanoic acid | 1:1                 | MD 1:1    | 0.941 ± 0.022       | 9.2                          | 0.868                        | 0.870 [25,28]                 |
| DL-Menthol | Dodecanoic acid | 2:1                 | MD 2:1    | 0.897 ± 0.018       | 12.3                         | 0.897                        | 0.894 [25]                    |
| DL-Menthol | Dodecanoic acid | 3:1                 | MD 3:1    | 0.903 ± 0.029       | 12.9                         | 0.870                        | 0.860 [33]                    |
| Thymol     | Dodecanoic acid | 1.25:1              | TD 1.25:1 | 0.832 ± 0.018       | 23.2                         | 0.894                        | 0.900 [73]                    |
| DL-Menthol | Thymol          | 1:1                 | TM 1:1    | 0.796 ± 0.023       | –                            | 0.951                        | 0.937 [72]                    |
| DL-Menthol | Palmitic acid   | 12:1                | MP 12:1   | 0.901 ± 0.035       | 22.5                         | 0.914                        | 0.890 [31]                    |

\*Experimental data from literature.

Following the vacuum drying process, the water content in the ESs was measured using a Karl Fischer Titrator (Metrohm India Ltd.), while density was determined with a specific gravity analyzer (Pycnometer, Eisco Labs). Additionally, Nuclear magnetic resonance (NMR) (400 MHz YH; JEOL, USA) spectroscopy and Fourier-Transform Infrared Spectroscopy (FT-IR, Frontier, Perkin Elmer, USA) were conducted. Furthermore, Differential Scanning Calorimetry (DSC, Discovery DSC 25; Waters, USA) experiments were performed wherein the samples were equilibrated at 40 °C, followed by a temperature scan cycle ranging from –60 °C to 60 °C with a rate of 5 °C/min, under inert atmosphere [43].

### 2.3. Phase behavior study of the water-eutectic solvent binary system

The hydrophobic behaviour of the ESs was studied through binary liquid–liquid equilibria (LLE) experiments. In each experiment, 1 ml of ES was thoroughly mixed with 1 ml of water in a vial using a vortex mixture. The vial was placed in a water bath at 298.15 K and left overnight for phase separation. After this, Karl Fischer titration was carried out to determine the water content in the upper (ES phase) and lower (aqueous phase) layers. <sup>1</sup>H NMR spectroscopy was also carried out for composition analysis of upper and lower layers. The samples drawn from each layer were mixed with 0.5 ml of DMSO- d<sub>6</sub> in an NMR tube (Sigma-Aldrich) for <sup>1</sup>H analysis. Eq. (1) was utilized to determine the mole fraction of each component within a mixture based on <sup>1</sup>H NMR.

$$x_i = H_i / \sum_{i=1}^n H_i \quad (1)$$

Where  $H_i$  represents the peak area attributed to a single hydrogen atom of the  $i^{\text{th}}$  component, while  $x_i$  denotes the mole fraction of that component within the mixture.

### 2.4. Ternary liquid–liquid equilibria study of 1,3-PDO aqueous solution using eutectic solvent

In each liquid–liquid extraction (LLE) experiment, an aqueous solution was prepared by mixing 2 ml of 1,3-PDO with 8 ml of water, with the concentration of 1,3-PDO set at 20 vol% to mimic its presence in the fermentation broth closely. 10 ml of ES as solvent was mixed with the prepared aqueous solution in a 50 ml glass set up to perform the extraction experiment, designed to prevent solvent loss and maintain temperature (Fig. S2). The resulting mixture was stirred at 500 rpm for 6 h at a temperature of 298.15 K ( $\pm 1$  K) under atmospheric pressure. After the stirring period, agitation ceased, and the mixture was allowed to settle under static conditions for 24 h at 298.15 K. Following a settling period, samples were drawn from both the extract (ES-rich phase) and raffinate (water-rich phase) phases using syringes. Approximately 0.1 ml of each sample was loaded into NMR tubes and dissolved in 0.5 ml of DMSO-*d*<sub>6</sub> via vortex mixing, and subsequent <sup>1</sup>H analysis was carried out using NMR for composition analysis. Water content in each phase was determined using the Karl Fischer titrator. The experiments were conducted with a minimum of three repetitions. For each measurement, the distribution coefficient ( $\beta$ ) and selectivity ( $S$ ) were calculated based on

the composition of the components in the mixture. These parameters were determined using Eq. (2) and Eq. (3).

$$\beta = x_{1,3-PDO}^E / x_{1,3-PDO}^R \quad (2)$$

$$S = \frac{x_{1,3-PDO}^E / x_{1,3-PDO}^R}{x_{water}^E / x_{water}^R} \quad (3)$$

Here,  $x_{1,3-PDO}^E$  and  $x_{1,3-PDO}^R$  represent the mole fractions of the 1,3-PDO while  $x_{water}^E$  and  $x_{water}^R$  denote the mole fractions of the water in the extract and raffinate phases, respectively. Further, the two phases (ES phase and aqueous phase) were separated, and the %extraction efficiency was calculated as the ratio of the mass of 1,3-PDO extracted to the total mass of 1,3-PDO initially present in the aqueous phase, multiplied by 100.

## 3. Computational methodology

### 3.1. Force field

The OPLS-AA (optimized potentials for liquid simulations-all atom) force field was used to simulate solvents and 1,3-PDO by evaluating the overall energy by combining individual intra- and intermolecular energy contributions [44].

$$E_{\text{Bonds}} = \sum_{\text{bonds}} K_{b,i} (r_i - r_{0,i})^2 \quad (4)$$

$$E_{\text{Angles}} = \sum_{\text{Angles}} K_{b,i} (\theta_i - \theta_{0,i})^2 \quad (5)$$

$$E_{\text{Dihedrals}} = \sum_{\text{Dihedrals}} \left[ \frac{1}{2} V_{1,i} (1 + \cos [\text{CSLEF}]_i) + \frac{1}{2} V_{2,i} (1 - \cos 2[\text{CSLEF}]_i) + \frac{1}{2} V_{3,i} (1 + \cos 3[\text{CSLEF}]_i) + \frac{1}{2} V_{4,i} (1 - \cos 4[\text{CSLEF}]_i) \right] \quad (6)$$

$$E_{\text{Nonbonded}} = \sum_i \sum_{j>i} \left\{ \frac{q_i q_j e^2}{r_{ij}} + 4 \epsilon_{ij} \left[ \left( \frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left( \frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] \right\} \quad (7)$$

$$E_{\text{Total}} = E_{\text{Bonds}} + E_{\text{Angles}} + E_{\text{Dihedrals}} + E_{\text{Nonbonded}} \quad (8)$$

Eq. (8) outlines the total energy in the OPLS-AA force field, while Equations (4–6) establish relationships for distances ( $r_i$ ) and angles ( $\theta_i$ ) relative to their equilibrium values ( $r_{0,i}$  and  $\theta_{0,i}$ ), with  $K$  representing the force constant and  $V$  the Fourier coefficient for dihedrals. Eq. (7) describes total nonbonded interactions between the atoms: Coulomb's law governs electrostatic energy, considering partial atomic charges ( $q_i q_j$ ), interatomic separation ( $r_{ij}$ ), and Van der Waals interactions were determined by the 12–6 Lennard-Jones potential based on interatomic separation ( $r_{ij}$ ), energy well depth ( $\epsilon$ ), and interatomic separation at zero potential ( $\sigma$ ). Standard combination rules,  $\epsilon_{ij} = (\epsilon_{ii} \epsilon_{jj})^{1/2}$  and  $\sigma_{ij} = (\sigma_{ii} \sigma_{jj})^{1/2}$ , were used to calculate LJ parameters for different atom types.

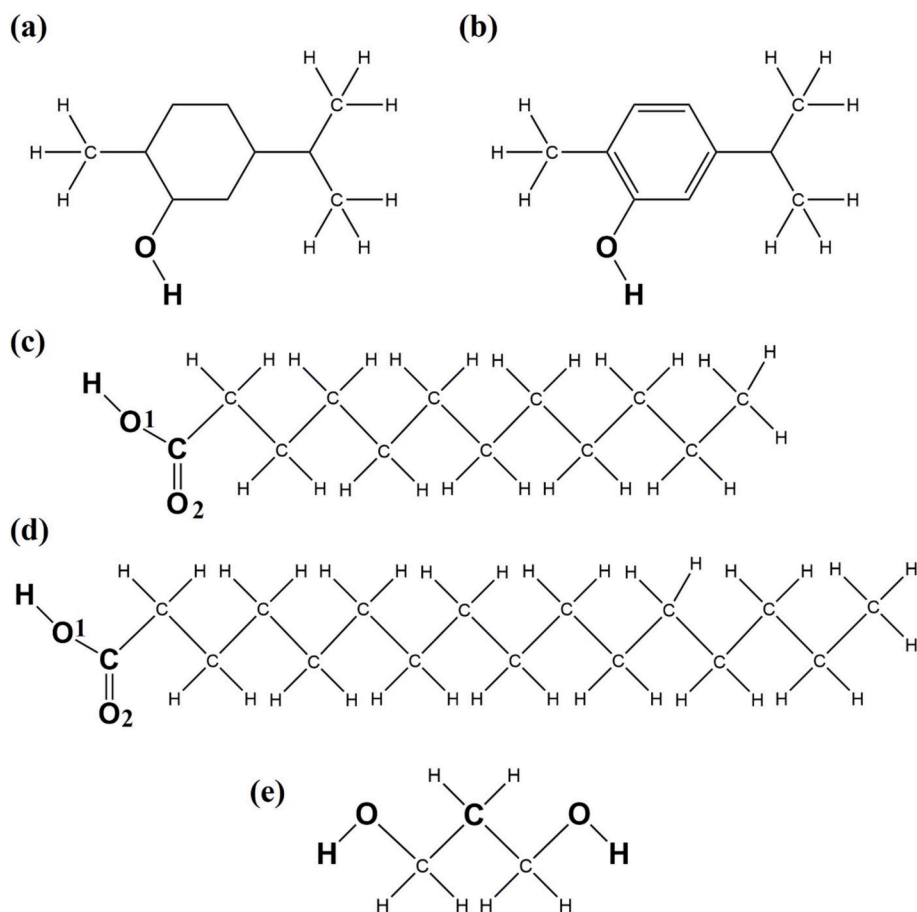
The force field parameters were assigned using the Ligpargen web

server, which employs 1.14\*CM1A charge models [45]. This tool parameterizes small organic molecules within the OPLS-AA force field framework [46,47]. The water model considered in the present work is OPC3 (3-point Optimal Point Charge) [48,49]. The OPC3 model was selected because it provides the best agreement with the experimental values among the available 3-point point water models [50]. All the force field parameters are available in the [supplementary information \(Supplementary 1\)](#).

### 3.2. Molecular dynamics simulation Details

Molecular dynamics (MD) simulations of ES-water binary systems and ES-water-PDO ternary systems were carried out to study the microscopic and macroscopic behaviour of the systems. Fig. 1 shows the molecular structures of the compounds along with their atom types studied in the present work. For ES/Water binary systems, the number of ES molecules was kept to 150, whereas the number of water molecules was adjusted to maintain an ES/Water volume ratio close to the experimental conditions (ES/Water volume ratio = 1:1). For ES/1,3-PDO/Water ternary systems, the number of ES molecules were also kept to 150, while the molecules of water and 1,3-PDO were adjusted to achieve a 20 vol% PDO in the aqueous phase and a ES/aqueous phase volume ratio close to 1:1 to mimic the experimental conditions. Table S1 summarizes the molecules of ESs, 1,3-PDO, and water molecules utilized in the present study. A three-box strategy was employed to pack the molecules for binary and ternary system simulations. The DES was placed in the middle, while water or the aqueous phase (water + 1,3-PDO) was placed on both sides. The initial configurations of the various systems were prepared using the PACKMOL software [51].

All MD simulations were carried out using the LAMMPS package (version LAMMPS-64bit-8Feb2023.exe) [52]. It was conducted in a three-dimensional (3D) space, ensuring periodic boundary conditions in all dimensions. Intermolecular interactions were represented using the Lennard-Jones (LJ) potential with a cutoff distance of 12.0 Å, along with analytical tail correction. The PPPM (Particle-Particle Particle-Mesh) method was used to calculate the long-range electrostatic interactions [53]. At first, the system underwent energy minimization via the steepest descent algorithm to refine its initial configuration. Subsequently, initial velocities were assigned to all atoms from a Gaussian distribution at the specified temperature to ensure a realistic starting point. The simulation was initiated with a base temperature of 5 K, progressively elevating the box temperatures to 10 K, 50 K, 100 K, 150 K, 200 K, 300 K, and finally 350 K. Employing a timestep of 0.5 fs, the system underwent simulation for 1 ns at each temperature increment. Following successful equilibration at 350 K, a gradual cooling process was employed to reduce the temperature to 298.15 K. The temperature annealing process was adopted to facilitate proper mixing of the viscous systems. A 50 ns production run was subsequently conducted under NPT ensemble conditions. System dynamics were simulated by integrating equations of motion using the velocity-Verlet algorithm, employing a timestep of 1.0 fs (fs). Temperature and pressure were controlled throughout the NPT simulation using the Nose-Hoover thermostat [54,55] and Parinello-Rahman barostat [56]. The thermostat maintained a constant temperature by adjusting atom velocities, while the barostat upheld pressure at 1 atm through adjustments to the simulation box volume. Radial distribution functions (RDF) and spatial distribution functions (SDF) were extensively examined to gain insights into molecule interactions within the system. Specifically, the TRAVIS package



**Fig. 1.** Chemical structure of the components, accompanied by their respective atom types: (a) DL  $\pm$  Menthol, (b) Thymol, (c) Dodecanoic acid, (d) Palmitic acid, and (e) 1,3-Propanediol.



was employed for SDF analysis [57], with an appropriate *iso*-value selected to visualize spatial distributions in three dimensions. Additionally, hydrogen bonding analysis using the Visual Molecular Dynamics (VMD) tool [58] provided insights into the nature and strength of interactions between solvents and 1,3-PDO, offering valuable information on interactions. An additional 50 ns NVE simulation was done to evaluate mean square displacement (MSD) and self-diffusion coefficient, contributing to understanding particle dynamics and system behavior [59].

### 3.3. COSMO-SAC calculations

Utilizing the quantum chemical-based conductor-like screening model-segment activity coefficient (COSMO-SAC), the infinite dilution activity coefficient of 1,3-PDO in ESs was predicted [60–62]. The computational process involved several key steps, starting with drawing molecular structures using Avogadro (version 1.2.0) [63]. Subsequently, these structures were optimized using the Gaussian16 computational package with the B3LYP theory and 6–311++g (d, p) basis set, followed by frequency calculations to ensure the structures corresponded to energy minima [64]. COSMO files were then generated for each molecule using Gaussian16 with the BVP86/TZVP/DGA1 level of theory and basis set, integrating the “scrf=COSMORS” keyword to indicate the use of the COSMO solvation model [65,66]. The generated COSMO files were inputs for the COSMO-SAC package, open-source software that computed sigma profiles [67]. These sigma profiles were then utilized as inputs again for the COSMO-SAC package, facilitating the calculation of the infinite dilution activity coefficient of 1,3-PDO in the investigated ESs using the model of Hsieh et al. as implemented in the package [68].

## 4. Results and Discussions

### 4.1. Synthesis and characterization of eutectic solvents

The synthesized eutectic solvents were transparent and appeared as clear liquids. The water content, determined via Karl Fischer titration, was below 1 wt% for all prepared ESs (Table 1). Also, as illustrated in Figs. S3–S6, 1H NMR analysis confirmed the synthesis of ESs [41] and the composition of the HBA and HBD in the required ratios, as corroborated by the algebraic method outlined in Tables S2–S5 [31,69]. The FT-IR spectrum presented in Fig. S7 confirms eutectic mixture formation. The C–H and C–C vibrations remained unchanged, indicating that the HBA and HBD were not degraded in the ESs. The pronounced O–H peaks in the ESs at 3200–3500 cm<sup>−1</sup> depict the stretching vibrations of the hydroxyl group (–OH), appearing relatively broader than those of the individual HBA and HBD [70]. Additionally, the peaks in the range of 2850–2970 cm<sup>−1</sup> corresponding to a sp<sup>3</sup> C–H bond further reveal the formation of ESs [71].

Differential scanning calorimetry (DSC) is a widely utilized thermal analysis method crucial for measuring the energy and temperature

associated with various thermal events, such as glass transitions, crystallization, and melting. In our study, DSC analysis was conducted to evaluate thermal events, mainly focusing on the melting point depression of ES compared to its components (Fig. S8). This depression serves as a robust indicator of successful ES formation. Our findings suggest that the hydrogen bond interactions within HBA and HBD significantly contribute to the melting point depression rather than any physical or structural disparities between the HBA and HBD. For instance, two distinct melting points were observed in the case of the DL ± Menthol: Dodecanoic acid ES. This phenomenon is attributed to the unique thermal characteristics of menthol, which exhibits two melting points (at 34 °C and 27 °C), as revealed by DSC analysis [69]. Furthermore, our investigation into thymol and menthol-based DES, along with their liquid–liquid equilibria, aimed to elucidate the non-ideal behavior of hydrophobic DESs, which exhibit the widest temperature range and can maintain their liquid state across a broad range of temperatures [72]. The observed peaks in the thermograms of ESs are consistent with previous research findings. Moreover, our study indicates the potential to establish an extensive range of operational temperatures within which these solvents remain liquid, facilitating their utilization at ambient temperature levels.

### 4.2. Force field validation

Before commencing the simulation, estimating the density of the pure components involved in the present work was imperative to validate the force field parameters. Subsequently, these estimates were juxtaposed with experimental data for comparison. Table 2 presents the simulated density values and corresponding experimental densities for various ESs. The simulated and experimental densities of pure systems were compared using charge scaling factors of ± 0.8 and ± 1.0. The analysis showed that the ± 0.8 scaling factor produced a low relative deviation (AARD of 0.858 %), whereas the ± 1.0 scaling factor resulted in a slightly higher relative deviation (AARD of 2.268 %). These findings underscore the superior accuracy of the ± 0.8 scaling factor in estimating densities compared to the ± 1.0 scaling factor for the components in our study. Notably, the absolute relative deviation for 1,3-PDO and pure ESs remained below 2 % with the ± 0.8 charge scaling factor.

Further, the mean squared displacement (MSD) was employed to calculate the self-diffusion coefficient of the components, including the HBA and HBD in the ES. The calculated self-diffusion coefficients were compared with values reported in the literature to validate the results. The MSD is defined as:

$$MSD = \frac{1}{N} \left[ \sum_{i=1}^N |r_i(t) - r_i(0)|^2 \right] \quad (9)$$

The self-diffusion coefficient (D) is derived from the slope of MSD over time, expressed as:

**Table 2**  
Experimental and Calculated liquid densities (g/cm<sup>3</sup>) at 298.15 K for ESs and 1,3-PDO.

| Molecules | Exp. Density (g·cm <sup>−3</sup> ) | Simulated density (g·cm <sup>−3</sup> ) [±0.8*Charge] | %ARD* | Simulated density (g·cm <sup>−3</sup> ) [±1.0*Charge] | %ARD* |
|-----------|------------------------------------|-------------------------------------------------------|-------|-------------------------------------------------------|-------|
| MD 1:1    | 0.890                              | 0.897 ± 0.002                                         | 0.786 | 0.902 ± 0.001                                         | 1.348 |
| MD 2:1    | 0.893                              | 0.894 ± 0.001                                         | 0.112 | 0.887 ± 0.002                                         | 1.115 |
| MD 3:1    | 0.893                              | 0.896 ± 0.002                                         | 0.336 | 0.908 ± 0.002                                         | 1.679 |
| TD 1.25:1 | 0.894                              | 0.905 ± 0.001                                         | 1.297 | 0.912 ± 0.001                                         | 2.013 |
| TM 1:1    | 0.951                              | 0.936 ± 0.001                                         | 1.577 | 0.935 ± 0.001                                         | 1.587 |
| MP 12:1   | 0.914                              | 0.920 ± 0.002                                         | 0.875 | 0.941 ± 0.002                                         | 2.954 |
| 1,3-PDO   | 1.049 [74]                         | 1.032 ± 0.001                                         | 1.681 | 0.963 ± 0.001                                         | 8.250 |
| Water     | 1.000                              | 0.998 ± 0.001                                         | 0.200 | 0.998 ± 0.001                                         | 0.200 |
| %AARD     |                                    |                                                       | 0.858 |                                                       | 2.268 |

\*Absolute Relative Deviation (ARD) =  $\frac{|\text{Exp. Density} - \text{Simulated Density}|}{\text{Exp. Density}} \times 100$

$$D_s = \frac{1}{6} \lim_{n \rightarrow \infty} \frac{d}{dt} \left( \sum_{i=1}^N |r_i(t) - r_i(0)|^2 \right) \quad (10)$$

Where  $r(t)$  and  $r(0)$  indicate the position of the atom 'i' center of mass at time  $t$  and time  $0$ ; the total encompasses all species; and the brackets denote the ensemble average. The three-dimensional system is accounted for by the  $1/6$  factor and the number of particles,  $N$ . Further, after determining that the  $\pm 0.8$  scaling factor provides accurate density estimates, this scaling factor was subsequently applied to calculate the self-diffusion coefficient for validating the results, as presented in Table S6 of the [supplementary material](#). The analysis revealed a percentage error ranging from 1 % to 6.5 %, indicating that the  $\pm 0.8$

scaling factor performs well for density estimation and offers reliable predictions for the self-diffusion coefficient of the components studied. Therefore, a 0.8-scaled charge model was employed for the rest of the study.

#### 4.3. Phase equilibria analysis of ES-water binary system

ES-water binary systems have been studied with two main objectives: (a) assessing the stability of ESs in the presence of water and (b) evaluating the hydrophobicity of ESs. The  $^1\text{H}$  NMR analysis of the raffinate phase of the binary systems was conducted to detect even trace amounts of starting components of ESs. All  $^1\text{H}$  NMR spectra showed

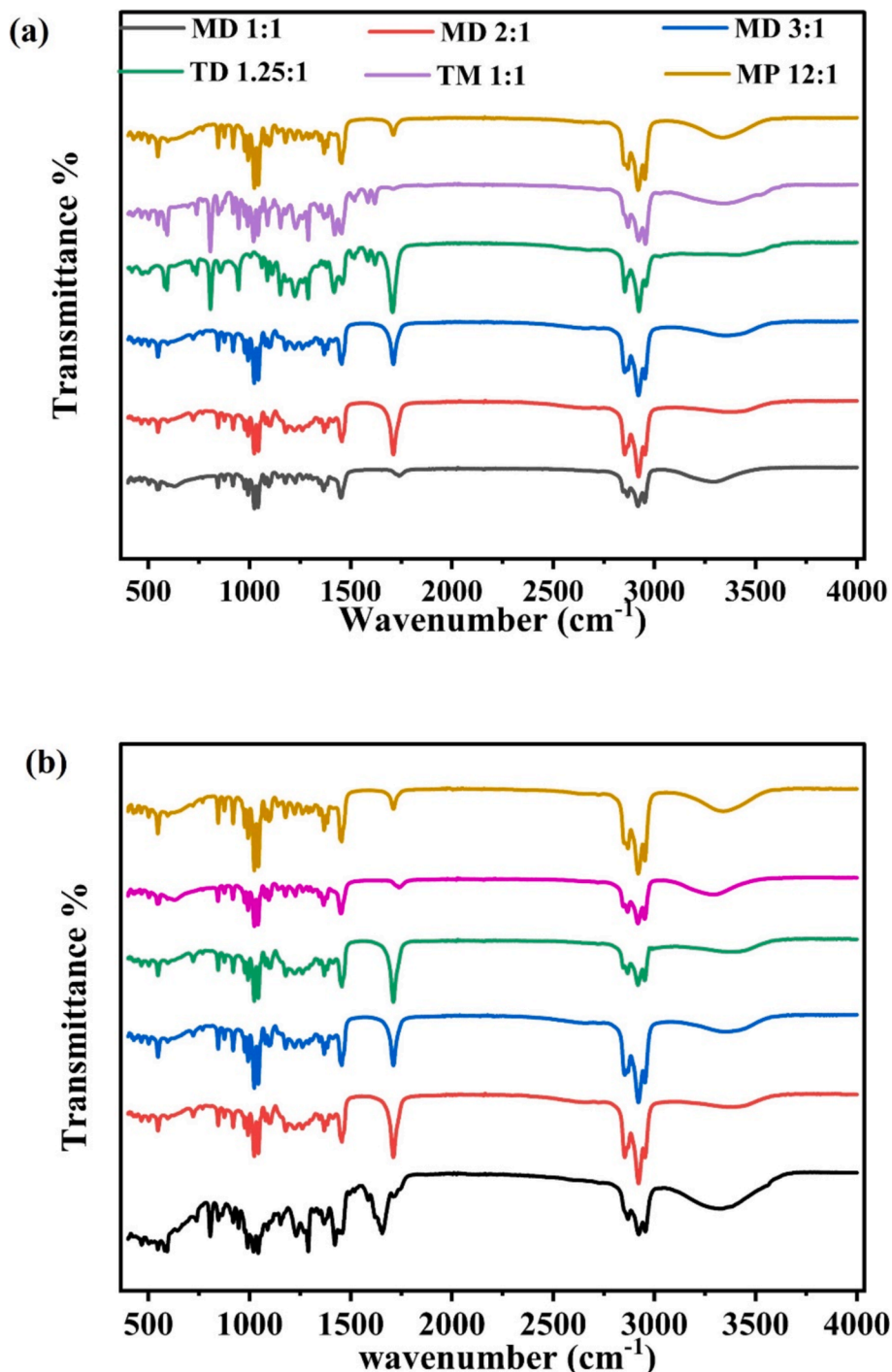


Fig. 2. FT-IR Spectra of ESs (a) Before and (b) After Adding Water for Phase Equilibria Study.

peaks corresponding only to water, with no other peaks observed (Figs. S9-S12), which indicates the stability of synthesized hydrophobic ESs in the presence of water. Consequently, these hydrophobic ESs may be suitable for recovering components without loss to the aqueous media, thereby avoiding associated environmental and economic costs. The hydrophobic behaviour of the ESs was assessed by determining the water content in the extract phase (ES phase) through Karl Fischer titration. Table S7 shows the wt% of water in the ES phase after saturating the ES with water. The water content in ESs is limited to low values, ranging from 1.11 to 3.18 wt%. DL-menthol: dodecanoic acid (MD 3:1) shows the most hydrophobic behaviour with a water content of only 1.11 wt%, whereas thymol: dodecanoic acid (TD 1.25:1) has the highest water content of 3.18 wt%. While these ESs are hydrophobic, they still absorb some amounts of water. However, this absorption is minimal and does not affect the structural and chemical stability of the ESs, as evident by both the  $^1\text{H}$  NMR (Figs. S9-S12) and FTIR spectra

(Fig. 2). The FTIR spectra indicate no structural changes or significant shifts in the FTIR spectra of ESs after exposure to water, highlighting the stability of ESs in aqueous environments (Fig. 2). These findings substantiate the suitability of the DES for further analysis in ternary systems for the extraction of 1,3-PDO, as it enhances the selectivity of separation, ensuring efficient and effective extraction processes.

The interfacial behaviour and mutual solubility of ESs and water were studied using molecular dynamics simulations. The density profiles of the water-ES binary systems are shown in Fig. 3. These figures clearly show well-defined aqueous and solvent phases with distinct interfacial regions. Notably, no DES molecules were observed within the aqueous phase, while a few water molecules entered the solvent phase. These observations are consistent with the experimental observations. The composition of water in the solvent phase was estimated using the following relation:

$$\text{Mole fraction of water in the solvent phase} = \frac{\text{number of water molecules in bulk and interface}}{\text{number of water molecules in bulk and interface} + \text{number of solvent molecules}} \quad (11)$$

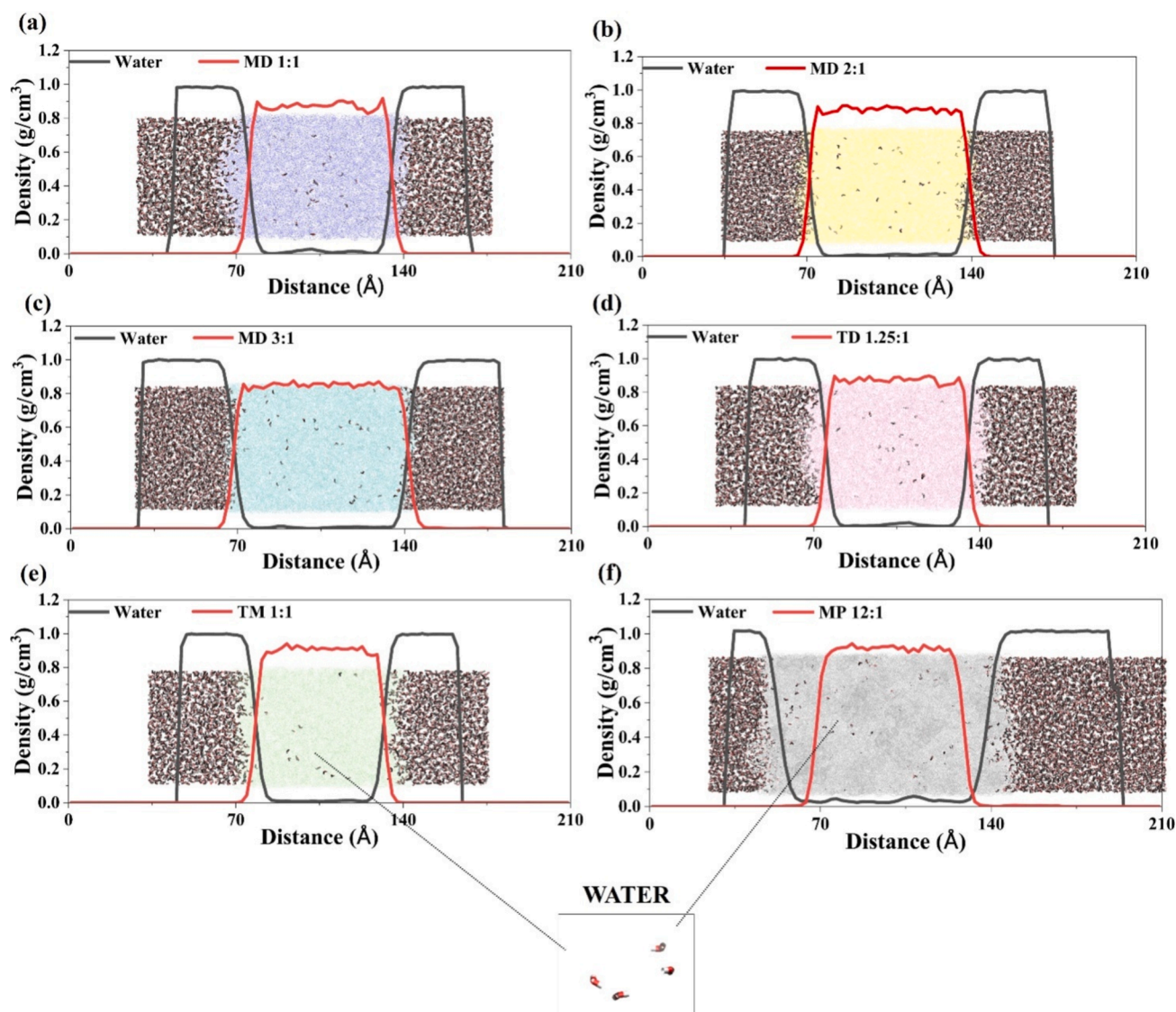


Fig. 3. The mass density profiles of the ES-water binary system obtained after a 40 ns simulation run at 298.15 K. (a) MD 1:1, (b) MD 2:1, (c) MD 3:1, (d) TD 1.25:1, (e) TM 1:1, and (f) MP 12:1.



Table S8 shows the distribution of water molecules between the water phase and ES phase after 40 ns of simulation at 298.15 K and 1 atm pressure. It is clear from the table that a small number of water molecules have moved from the aqueous phase to the solvent phase in each binary system. The calculated mole fraction of water in the solvent phase varies from 0.133 to 0.49. The equivalent composition of water in terms of wt% in the solvent phase varies from 0.411 % (DL-menthol: dodecanoic acid (MD 3:1)) to 2.2 % (thymol: dodecanoic acid (TD 1.25:1)). The composition of water in solvent phase in terms of wt% from molecular dynamics follows the same trend as observed from the experimental data using Karl Fischer titration. It is also evident that molecular simulation underpredicts the water composition in the solvent phase in each case (Fig. S13). Fig. S14 shows the volume slice plot of the binary systems, illustrating the density distribution of water molecules around the HBA and HBD components of ESs. This visualization highlights areas of high water concentration, revealing interaction patterns within the system. In TD 1.25:1, high water density is observed around both HBA and HBD components, whereas in menthol-based ESs, low water density is observed around menthol. These indicate that menthol is the primary component responsible for the hydrophobic behaviour of the studied ESs. Based on the study of binary ES-water systems, the ESs exhibit the following order of hydrophobic behaviour, from the greatest to least: MD 3:1 > MD 2:1 > MP 12:1 > MD 1:1 > TM 1:1 > TD 1.25:1. Overall, all the studied ESs exhibit excellent structural stability and hydrophobic behaviour, making them potentially useful for the recovery of compounds from the aqueous phase.

In response to the observed superior performance of MD 3:1, a stability study was conducted for this eutectic solvent in a binary system with water at a 1:1 vol ratio. The mixture was maintained at 3–5 °C, resulting in phase separation due to the eutectic solvent's melting point of 12.9 °C. After each cycle, the solvent and water phases were separated, and the recovered ES was allowed to melt at room temperature before being re-mixed with water in the same volume ratio. This process was repeated for five cycles (Fig. S15). The ES was analyzed in each cycle using a Karl Fischer titrator to measure its water absorption capacity. The results indicate that the water content remained consistent across all cycles, ranging from 1.11 to 1.23 wt% (Table S9). Additionally, <sup>1</sup>H NMR analysis conducted after the fifth cycle, as shown in Fig. S16, confirmed the solvent's chemical integrity, with no detectable degradation or loss of components. These results highlight the robustness and reusability of MD 3:1, making it a promising candidate for industrial extraction processes.

#### 4.4. Phase equilibria analysis of ES-1,3-PDO-water ternary system

After performing the ES-water binary equilibria study, ES-1,3-PDO-water ternary systems were studied. The <sup>1</sup>H NMR spectra peaks were analyzed to estimate mole fractions of ES and 1,3-PDO in each phase, while Karl Fischer titration was used to quantify water. 1,3-PDO has three different types of H-atoms, resulting in three distinct NMR peaks: one at 1.56 ppm due to the central –CH<sub>2</sub>– group, a second at 3.45 ppm due to the –CH<sub>2</sub>– group attached with –OH group and a third peak at 4.41 ppm due to –OH group. The peak at 3.45 ppm was used to quantify 1,3-PDO in both phases (Fig. S17). In DL-menthol: dodecanoic acid-

based ternary systems (MD 1:1, MD 2:1, and MD 3:1), peak at ~ 3.15 ppm, corresponding to a single H-atom (H<sub>9</sub>), was used to quantify ES in extract phase (Figs. S18–S20). For the thymol: dodecanoic acid (TD 1.25:1) based ternary system, the peak at 0.85 ppm, due to the three hydrogens of decanoic acid (H<sub>5</sub>) was used for ES quantification in the extract phase (Fig. S21). In the thymol: menthol (TM 1:1) based ternary system, the peak at 6.57 ppm (H<sub>5</sub>) was used for ES quantification (Fig. S22), while the peak at ~ 0.84–0.86 ppm, corresponding to 24 hydrogens (H<sub>4</sub>) of palmitic acid, was used for quantification of MP 12:1 ES (Fig. S23).

<sup>1</sup>H NMR spectra of raffinate phases of all ternary systems are shown in Fig. S18 to Fig. S23, and all spectra clearly show no peak for the ES. NMR spectra are dominated by a water peak followed by a small peak of 1,3-PDO. These spectra indicate that ESs are also stable in equilibrium with an aqueous solution of 1,3-PDO and function efficiently as a solvent, eliminating the need for an extra solvent recovery step. Based on peak area and Karl Fischer titration, the mole fraction of components, distribution coefficient, and selectivity have been calculated and tabulated in Table 3. A sample calculation for estimating the composition of the components in the ternary system is shown in Table S10. The distribution coefficient quantifies how a solute is distributed between two immiscible phases at equilibrium. A high value implies that the solute can be efficiently transferred from one phase to another phase with minimal use of the solvent, making the process efficient and cost-effective. For all the ESs, the distribution coefficient is substantial, indicating their effectiveness for extracting 1,3-PDO from the aqueous solution. Among them, the menthol: dodecanoic acid ESs (MD 1:1, MD 2:1, and MD 3:1) are the most efficient, with distribution coefficient values ranging from 61 to 234, and of these, MD 3:1 is the best. Selectivity is another important separation parameter that measures how well one component can be separated from others. High selectivity leads to more efficient processes with higher purity of the desired component, reducing the need for downstream processes. For all the ESs, selectivity values are large, ranging from 480 to 18000, indicating excellent extraction ability of the ESs for 1,3-PDO compared to water. In terms of selectivity, MD 3:1 is the best ES. As a general trend, the more hydrophobic the ES, the higher the 1,3-PDO distribution coefficient and selectivity. Overall, distribution coefficients and selectivity follow the following order: MD 3:1 > MD 2:1 > MP 12:1 > MD 1:1 > TM 1:1 > TD 1.25:1.

Following the experimental results, we expanded our study to utilize molecular dynamics simulations to understand the separation mechanism of 1,3-PDO from an aqueous mixture. As outlined in Table S1, the number of molecules of the components was chosen to replicate the experimental conditions closely. Fig. 4 illustrates the local mass density profile of components after a 50 ns simulation run at 298.15 K. The 1,3-PDO is primarily concentrated in the ES phase or at the water-ES interface, indicating effective extraction of 1,3-PDO from the aqueous phase. This behavior is likely due to the favorable interactions between 1,3-PDO and the ES components, facilitating its preferential distribution into the ES phase. The water phase appears as a distinct cluster, effectively separating from the ES system. Moreover, the profiles indicate that 1,3-PDO facilitates water transfer from the aqueous phase to the ES phase, as supported by the Karl Fischer titration (Table S11). The

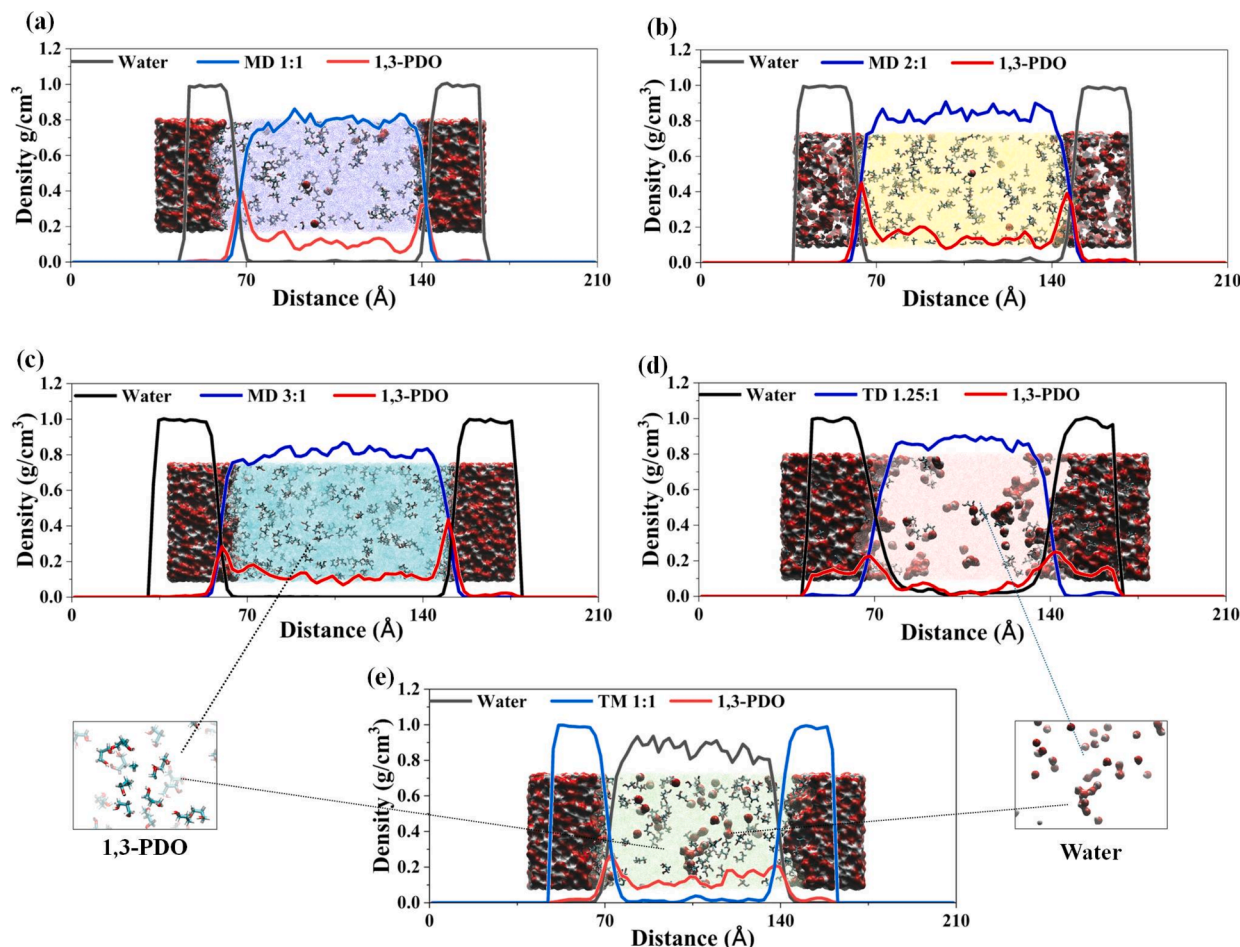
**Table 3**

Experimental composition (mass basis), distribution coefficient and selectivity for ES-1,3-PDO-water ternary system at  $T = 298.15$  K and  $p = 0.1$  Mpa.

| Solvent   | Extract phase |                 |               | Raffinate phase |                 |               | Distribution Co-efficient<br>$\beta_{1,3-PDO}$ | Distribution Co-efficient<br>$\beta_{water}$ | Selectivity<br>S |
|-----------|---------------|-----------------|---------------|-----------------|-----------------|---------------|------------------------------------------------|----------------------------------------------|------------------|
|           | $m_{ES}^E$    | $m_{1,3-PDO}^E$ | $m_{Water}^E$ | $m_{ES}^R$      | $m_{1,3-PDO}^R$ | $m_{Water}^R$ |                                                |                                              |                  |
| MD 1:1    | 0.7962        | 0.1780          | 0.0258        | 0.0000          | 0.0029          | 0.9971        | 61.38                                          | 0.0259                                       | 2372             |
| MD 2:1    | 0.7937        | 0.1848          | 0.0215        | 0.0000          | 0.0019          | 0.9981        | 97.26                                          | 0.0215                                       | 4523             |
| MD 3:1    | 0.7765        | 0.2105          | 0.0130        | 0.0000          | 0.0009          | 0.9991        | 233.89                                         | 0.0130                                       | 17,991           |
| TD 1.25:1 | 0.8314        | 0.1255          | 0.0431        | 0.0000          | 0.0060          | 0.9940        | 20.92                                          | 0.0434                                       | 482              |
| TM 1:1    | 0.7997        | 0.1702          | 0.0301        | 0.0000          | 0.0040          | 0.9960        | 42.55                                          | 0.0302                                       | 1409             |
| MP 12:1   | 0.9363        | 0.0385          | 0.0252        | 0.0000          | 0.0006          | 0.9994        | 64.17                                          | 0.0252                                       | 2546             |

Where, E represents Extract phase and R represents Raffinate phase.





**Fig. 4.** Mass density profile of the ES-1,3-PDO-water ternary system obtained after a 40 ns simulation run at 298.15 K. (a) MD 1:1, (b) MD 2:1, (c) MD 3:1, (d) TD 1.25:1, and (e) TM 1:1.

**Table 4**

Simulated composition (mass basis), distribution coefficient and selectivity for ES-1,3-PDO-water ternary system at  $T = 298.15$  K and  $p = 0.1$  Mpa.

| Systems   | Extract Phase |                 |               | Raffinate Phase |                 |               | $\beta_{1,3-PDO}$ | $\beta_{Water}$ | S       |
|-----------|---------------|-----------------|---------------|-----------------|-----------------|---------------|-------------------|-----------------|---------|
|           | $m_{ES}^E$    | $m_{1,3-PDO}^E$ | $m_{water}^E$ | $m_{ES}^R$      | $m_{1,3-PDO}^R$ | $m_{water}^R$ |                   |                 |         |
| MD 1:1    | 0.839         | 0.146           | 0.015         | 0.000           | 0.031           | 0.969         | 4.71              | 0.02            | 304.27  |
| MD 2:1    | 0.845         | 0.146           | 0.009         | 0.000           | 0.036           | 0.964         | 4.06              | 0.01            | 434.87  |
| MD 3:1    | 0.842         | 0.153           | 0.005         | 0.000           | 0.020           | 0.980         | 7.65              | 0.01            | 1499.40 |
| TD 1.25:1 | 0.897         | 0.081           | 0.022         | 0.000           | 0.130           | 0.870         | 0.62              | 0.03            | 24.52   |
| TM 1:1    | 0.840         | 0.142           | 0.018         | 0.000           | 0.038           | 0.962         | 3.74              | 0.02            | 199.88  |

Where, E represents Extract phase and R represents Raffinate phase.

distribution of components is detailed in Table S12-S13, with subsequent calculation of the distribution coefficient ( $\beta$ ) and selectivity (S) using mole fraction. Furthermore, the values of  $\beta$  and S on a mass basis are summarized in Table 4. The MD predicted composition of components in both phases in terms of mass fraction agrees well with the experimental results with a percent root mean square deviation (% RMSD) of 4.5 % only. Notably, MD simulation results exhibit a trend consistent with experimental findings. In general, menthol-based ESs exhibit a high distribution coefficient and selectivity for 1,3-PDO extraction, with MD 3:1 showing the best performance. The %extraction efficiency obtained experimentally for MD-based ES ranged from 96–99 %, highlighting their highly effective extraction capabilities (Table S14). In contrast, TD 1.25:1 exhibited significantly lower extraction efficiency, aligning with the results of  $\beta$  and S, as corroborated by molecular simulations and experimental analysis. These results emphasize the superior extraction performance of MD-based ESs

compared to TD-based ESs.

#### 4.5. Structural analysis

With the help of molecular dynamics simulations, the Radial Distribution Function (RDF) and Spatial Distribution Function (SDF) were estimated to understand the liquid–liquid extraction process of 1,3-PDO from aqueous systems employing ESs. These functions allowed us to discern the spatial arrangement and density distribution of molecules, thereby shedding light on the interactions between ESs and 1,3-PDO, as well as between water and 1,3-PDO. Further, Eq. (12) outlines the RDF function.

$$g(r) = \frac{\langle \rho(r) \rangle}{\rho} \quad (12)$$

Here,  $\rho(r)$  signifies the number density of particles at a distance “r” while

“ $\rho$ ” represents the overall number density of particles in the system. This equation offers insight into the likelihood of finding a particle at a specific distance from a reference particle, aiding in understanding particle distribution patterns and interactions within the system.

Additionally, the coordination number (CN) was determined by integrating the RDF function ( $g(r)$ ) up to the position of the first solvation shell, identified by the first local minimum in the RDF curve.

$$CN = 4\pi\rho \int_0^{R_{min}} r^2 g(r) dr \quad (13)$$

In this equation,  $4\pi r^2 dr$  denotes the volume element in a spherical shell, and  $\int_0^{R_{min}}$ , the integral is computed over the range of distances from 0 to  $R_{min}$ . This method quantitatively measures particle arrangement and interactions within the system. These RDF analyses provided valuable insights into the distribution of molecules in space and the frequency and proximity of interactions between specific atoms or groups in the ESs and 1,3-PDO. The analysis of the RDF and CN between ESs and 1,3-PDO, as illustrated in Fig. 5, highlights strong interactions across various ES compositions, reinforcing the notion of a robust connection between ESs and 1,3-PDO. In Fig. 5(a-c), the RDF and CN for menthol-dodecanoic ESs with varying ratios (1:1, 2:1, and 3:1) are presented, showing prominent interactions between the hydrogen atom attached to the -OH group of dodecanoic acid and the oxygen of the -OH group of 1,3-PDO, and the oxygen of DL-menthol with the hydrogen of 1,3-PDO. The CN values observed are 0.14, 0.17, and 0.20 for MD 1:1, MD 2:1, and MD 3:1, respectively. Fig. 5(d) shows the RDF of TD 1.25:1 with 1,3-PDO, revealing two series of CNs. The coordination number (CN(D)) for dodecanoic acid with 1,3-PDO is 0.07, while the coordination number (CN(T)) for thymol with 1,3-PDO is 0.10. These lower coordination numbers than other ESs indicate weaker interactions between the TD 1.25:1 ES and 1,3-PDO.

The RDF and SDF analysis of 1,3-PDO around water molecules in the ternary system were determined to gain further understanding (Fig. 6). RDF analysis reveals weak peaks and extremely low coordination numbers (CN) for the interaction between water and 1,3-PDO, indicating a minimal affinity between them. However, TD 1.25:1 (Fig. 6-(d)) stands out with a notably higher coordination number of approximately 0.36 with water compared to other menthol-based ESs, which exhibit significantly lower values. The sequence MD 3:1 > MD 2:1 > MD 1:1 > TM 1:1 > TD 1.25:1, observed in the RDF and coordination numbers between ES and 1,3-PDO, suggests that increasing menthol concentration in the ES enhances interactions with 1,3-PDO while diminishing interactions between water and 1,3-PDO. This phenomenon is likely due to reduced competition between ES components and water for hydrogen bonding with 1,3-PDO, allowing ES molecules to interact more effectively with 1,3-PDO. Thus, the RDF and coordination numbers between ES and 1,3-PDO inversely correlate with those between water and 1,3-PDO, highlighting the competitive nature of hydrogen bonding in these systems.

The SDF analysis was also carried out to obtain more detailed structural information on the molecular arrangement of 1,3-PDO around ESs and water at 298 K.15 (with the iso-value set at 1.5). These SDFs depicted the average density distribution of 1,3-PDO around ESs and water, as shown in Figs. 5 and 6, providing significant insights into their solvation behavior. The results obtained from the SDFs follow the same trend as those of the RDFs. The orange color in Figs. 5 and 6 represents the density distribution of 1,3-PDO around HBA/HBD and water molecules, respectively. Here, it is observed that the density distribution of 1,3-PDO is highest around menthol-dodecanoic acid-based ESs, such as MD 3:1, and lowest around TD 1.25:1. In Fig. 6, the SDF values reveal that higher densities of 1,3-PDO around water molecules are observed in compositions like TD 1.25:1. In comparison, lower densities are evident in compositions like MD 3:1. The trend observed from the combined analysis suggests that the distribution of 1,3-PDO molecules around water is influenced by the composition of the ES, with higher menthol

concentrations leading to reduced number densities while enhancing the density of 1,3-PDO around the HBA and HBD components. Moreover, high SDF values of 1,3-PDO around water molecules are found to minimize selectivity, thus making separation less efficient. Conversely, high SDF values of 1,3-PDO around the ES components enhance selectivity, easing the separation of 1,3-PDO from water.

#### 4.6. Hydrogen bond analysis

The number of hydrogen bonds involving 1,3-PDO, ESs, and water molecules was evaluated to gain insight into the interactions further. The quantification of the hydrogen bonds provides quantitative validation for observations made through RDF and SDF analyses. Molecules involved in hydrogen bonding were identified based on geometric criteria. A hydrogen bond was defined by a donor-acceptor distance cutoff of 3.5 Å and a hydrogen-donor-acceptor angle cutoff of 30° [75,76].

The bar graph in Fig. 7 highlights the average number of hydrogen bonds formed by ESs and water with 1,3-PDO molecules (Fig. S24). The bar graph indicates that certain ESs, particularly MD 3:1, exhibit a high number of hydrogen bonds with 1,3-PDO, suggesting a strong affinity between the two molecules. Menthol typically contains several functional groups, including hydroxyl (-OH) groups, while dodecanoic acid includes a carboxylic acid group (-COOH). When 1,3-PDO interacts with the ES, it's likely that hydrogen bonding will occur between the hydroxyl groups of menthol or the carboxylic acid group of dodecanoic acid and the hydroxyl groups of 1,3-PDO. Other menthol-based ESs also show significant interaction with 1,3-PDO. This strong interaction implies a high distribution coefficient for 1,3-PDO in these ESs. In water, the interactions of 1,3-PDO occur primarily via its hydroxyl groups (-OH). Water molecules can form hydrogen bonds with the hydroxyl groups of 1,3-PDO, where the oxygen atom of water acts as a hydrogen bond acceptor, while the hydrogen atoms of water act as hydrogen bond donors. In contrast, the number of hydrogen bonds between water and 1,3-PDO, especially in the TD 1.25:1 system, is notably high, indicating a strong affinity between 1,3-PDO and water. This suggests that 1,3-PDO has a higher affinity for water than the ES in this particular ES system. The high number of hydrogen bonds between ES and 1,3-PDO implies a strong affinity, which could lead to a high distribution coefficient. However, the high number of hydrogen bonds between water and 1,3-PDO, particularly in the TD 1.25:1 system, suggests that 1,3-PDO has a greater affinity for water, reducing the selectivity for separation using this ES.

#### 4.7. Self-diffusion coefficient

The transport properties, specifically the self-diffusion coefficient, were investigated to analyze the dynamics of 1,3-PDO in the ESs systems. Initially, the force field parameters for the ESs were validated, as presented in Table S6 of the supplementary information. Once the accuracy of the force field was confirmed, the focus shifted to studying the diffusion behavior of 1,3-PDO within the ternary systems. The MSD of 1,3-PDO and its self-diffusion coefficient (D) were calculated using Eqs. (9) and (10) by performing molecular dynamics simulations in the microcanonical ensemble for an additional 50 ns.

Fig. 8 shows that the MSD curves for 1,3-PDO in various ternary systems increase linearly over time. Among these systems, the lowest MSD slope is observed in MD 3:1, followed by MD 2:1 and MD 1:1, while the highest slopes are seen in TD 1.25:1 and TM 1:1. The difference in slopes can be linked to the hydrogen bonding between 1,3-PDO and the ESs. A lower slope suggests stronger hydrogen bonding, while a higher slope indicates weaker interactions. This pattern suggests that in systems like MD 3:1 and MD 2:1, where 1,3-PDO has higher solubility in the ES phase, the hydrogen bonding is stronger, reflected by the lower slope. On the other hand, in systems such as TD 1.25:1 and TM 1:1, where 1,3-PDO has a lower solubility in the ES phase, the hydrogen bonding is

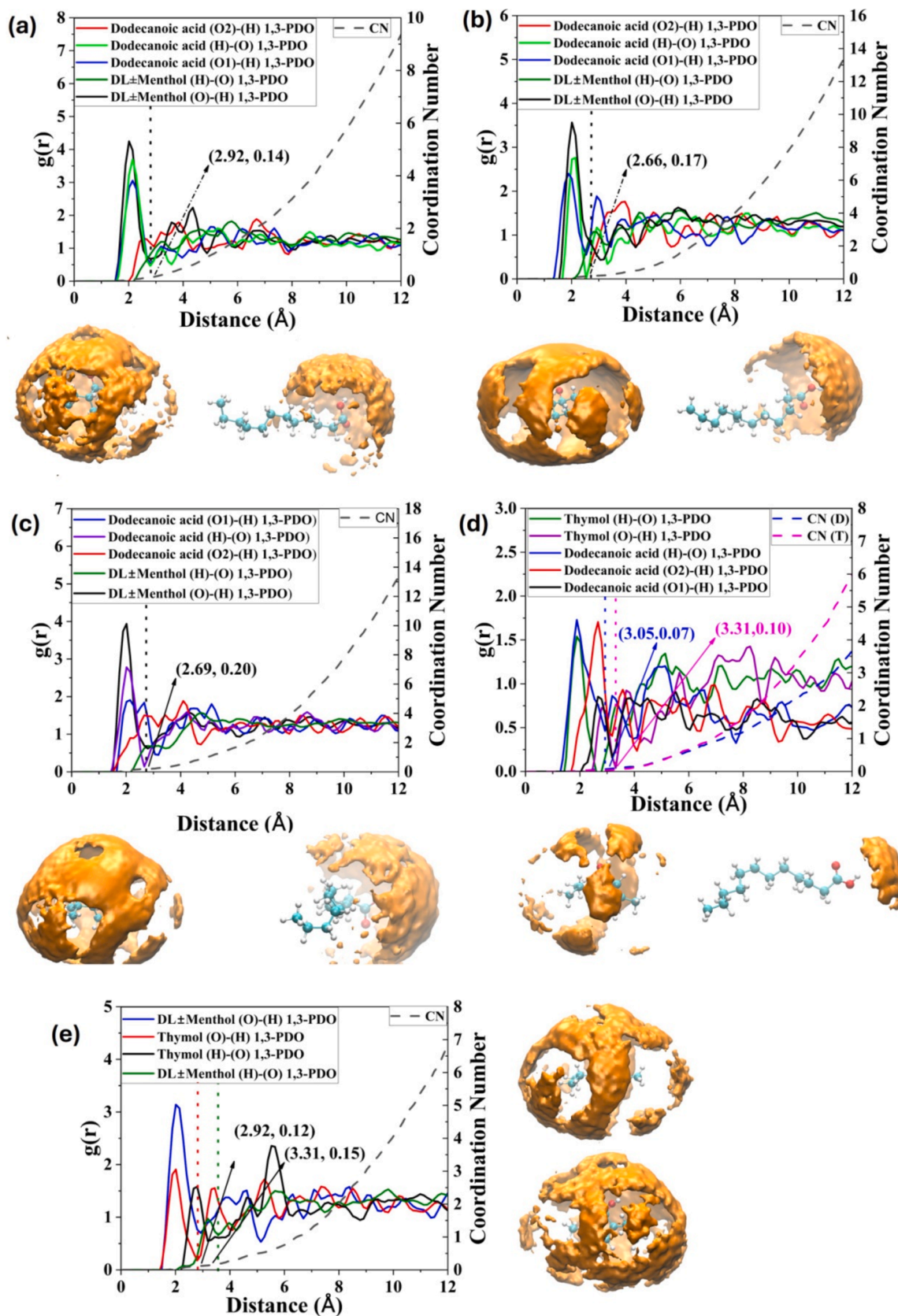


Fig. 5. RDF, CN and SDF of 1,3-PDO around ES, (HBA and HBD). (a) MD 1:1, (b) MD 2:1, (c) MD 3:1, (d) TD 1.25:1 and, (e) TM 1:1.



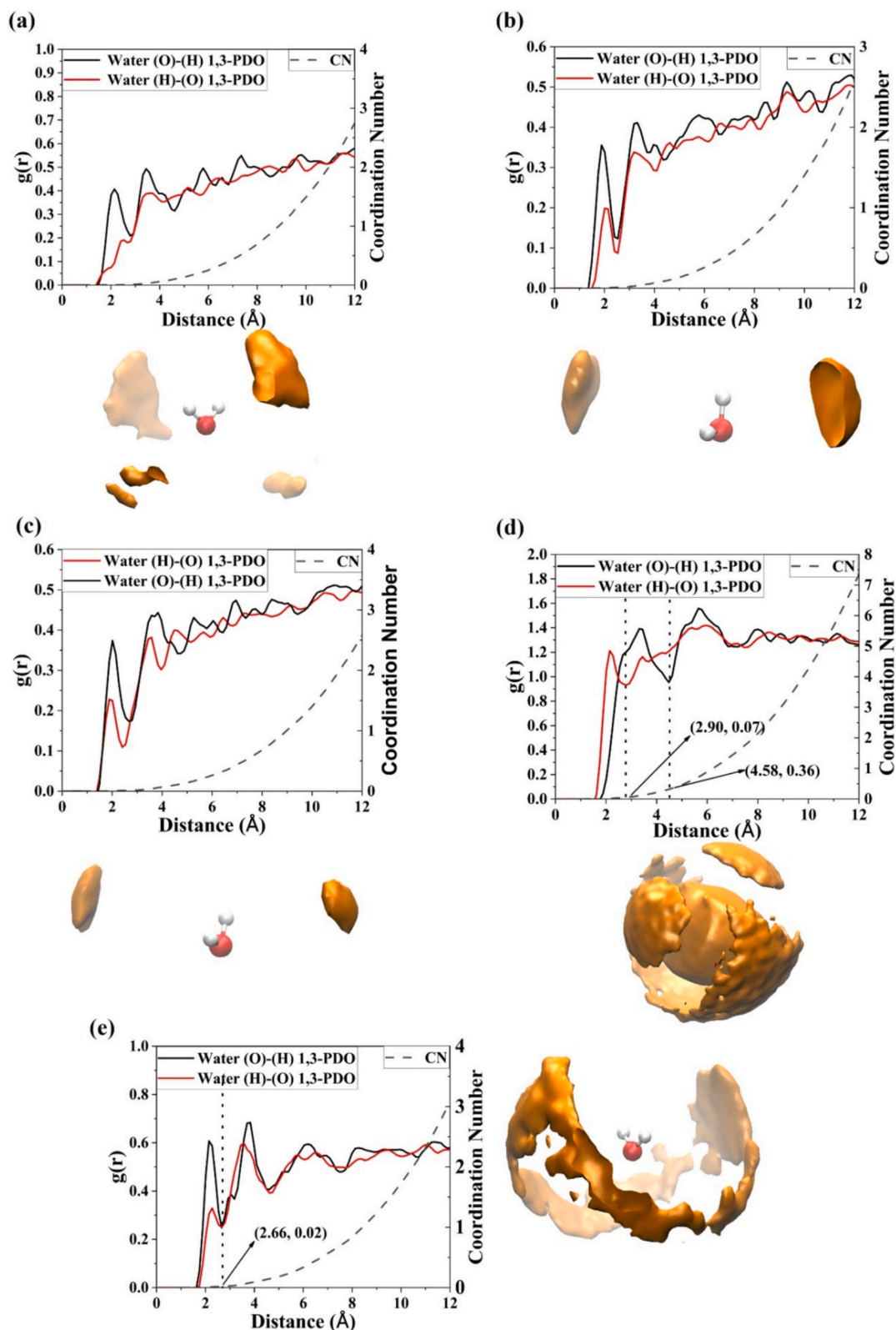


Fig. 6. RDF, CN and SDF of 1,3-PDO (Orange) around water (a) MD 1:1, (b) MD 2:1, (c) MD 3:1, (d) TD 1.25:1 and, (e) TM 1:1.

weaker, resulting in steeper MSD slopes. The  $D$  values were calculated for 1,3-PDO in the ES systems, and the results are shown in Table 5. The  $D$  value is less for MD 3:1 and high for TD 1.25:1. The lower the  $D$  value, the stronger the interaction of the molecular pair and the higher the solubility [77]. Menthol-dodecanoic acid ESs, especially MD 3:1, have the highest extraction efficiency compared to other studied ESs,

confirmed by the above analysis.

#### 4.8. COSMO-SAC Prediction

The sigma profiles of the individual components of ESs were calculated and combined in specific ratios to generate the final profiles used



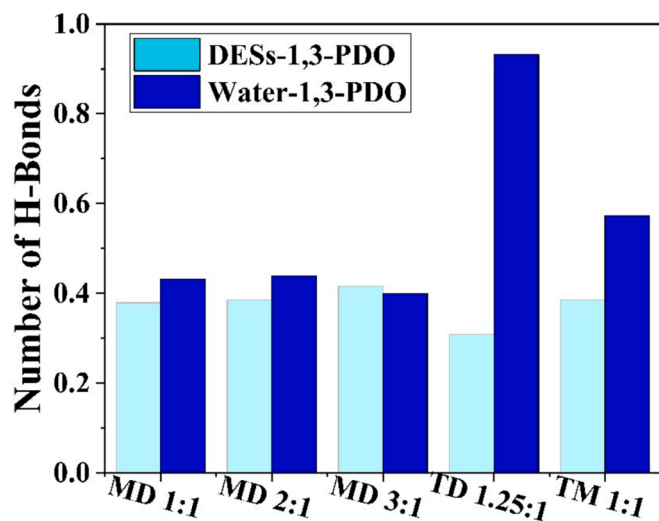


Fig. 7. Average number of H-bonds formed between ES and 1,3-PDO, and between water and 1,3-PDO, per 1,3-PDO molecule at 298.15 K as per molecular dynamics simulation.

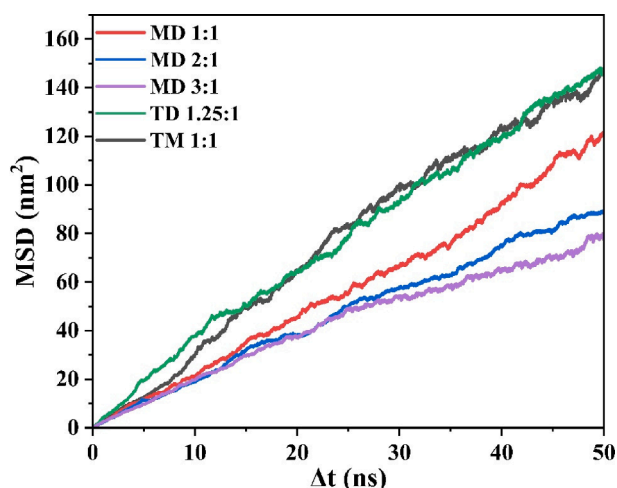


Fig. 8. MSD plot of 1,3-PDO in the ESs systems at 298.15 K.

Table 5

Diffusion coefficient of 1,3-PDO in ESs at 298.15 K.

| System    | Diffusion coefficient $\times 10^{-10} \text{ m}^2/\text{s}$ |
|-----------|--------------------------------------------------------------|
| MD 1:1    | $3.92 \pm 0.143$                                             |
| MD 2:1    | $3.48 \pm 0.323$                                             |
| MD 3:1    | $3.35 \pm 0.168$                                             |
| TD 1.25:1 | $4.85 \pm 0.252$                                             |
| TM 1:1    | $4.63 \pm 0.306$                                             |

in this study, aimed at understanding the potential interactions of ESs with 1,3-PDO, as illustrated in Fig. S25. The sigma profiles are categorized into three regions, which are the H-bond donor region ( $\sigma < -0.01 \text{ e}/\text{\AA}^2$ ), the non-polar region ( $\sigma$  between  $-0.01 \text{ e}/\text{\AA}^2$  and  $+0.01 \text{ e}/\text{\AA}^2$ ), and the H-bond acceptor region ( $\sigma > +0.01 \text{ e}/\text{\AA}^2$ ). The sigma profile of 1,3-PDO is symmetrical and broad. The peaks to the right side of  $+0.01 \text{ e}/\text{\AA}^2$  is due to the presence of oxygen atom indicating its ability to act as a hydrogen-bond acceptor. On the other hand, the hydroxyl hydrogen atoms presents a peak within the negative region specifying its ability to act as a hydrogen-bond donor. This reflects that the 1,3-PDO molecule has an ability to act both as a donor and an acceptor for hydrogen bonding. All ESs are showing significant non-polar regions, which are

attributed to the cyclohexane or aromatic rings and aliphatic chains in menthol, as well as the long alkyl tail in dodecanoic acid. As the menthol content increases, the solvent properties shift toward more hydrophobic and nonpolar characteristics. The sigma profiles of all ESs are distributed across the three regions. In the H-bond acceptor region, all ESs display similar profiles with limited spread. However, there is more spread in the H-bond donor region in menthol:dodecanoic acid ESs, while TM 1:1 and TD 1.25:1 exhibit comparatively narrower spreads. This suggests that MD-based ESs have a superior ability to act as H-bond donors compared to TM 1:1 and TD 1.25:1. From the analysis of the sigma profiles, it can be inferred that MD 3:1 are more non-polar or hydrophobic due to the presence of more menthol content. Additionally, the interactions between 1,3-PDO and MD are more favorable compared to those with TD 1.25:1 or other ESs.

The efficiency of ESs in extracting 1,3-PDO was assessed using the COSMO-SAC model, which predicts activity coefficient values to provide insights into the thermodynamic interactions between 1,3-PDO, water, and ES. These values indicate the strength of solute–solvent interactions, with lower values suggesting stronger interactions. The bar graph in Fig. 9 illustrates the infinite dilution activity coefficient values for 1,3-PDO and water across various ESs studied. The results show that 1,3-PDO exhibits varying degrees of interaction strength depending on the ES composition. Among the ES compositions studied, 1,3-PDO consistently exhibits lower activity coefficient values than water, indicating stronger solute–solvent interactions for 1,3-PDO within the ES systems. Notably, MD 3:1 ES demonstrates the strongest interactions with 1,3-PDO, characterized by its lowest infinite dilution activity coefficient. Increasing the menthol ratio in MD 3:1 enhances its affinity for 1,3-PDO, highlighting its potential effectiveness in extraction. In contrast, TD 1.25:1 shows a significantly higher infinite dilution activity coefficient, suggesting weaker interactions with 1,3-PDO. The strength of interactions for 1,3-PDO follows the sequence MD 3:1 > MD 2:1 > MD 1:1 > TM 1:1 > TD 1.25:1. From Fig. 9, it is observed that the infinite dilution activity coefficient values for water in ES are consistently higher compared to 1,3-PDO in ES across all compositions. This indicates weaker interactions between water and the ES systems studied, suggesting a less favorable water solvation environment than 1,3-PDO.

These results highlight that ES compositions affect solvation properties differently for water and 1,3-PDO. Specifically, while 1,3-PDO shows varying degrees of interaction strength depending on the ES composition, water consistently exhibits weaker interactions regardless of the ES formulation. By optimizing ES formulations to enhance interactions with 1,3-PDO while minimizing interactions with water, it is possible to improve the efficiency of extracting 1,3-PDO from aqueous systems using ESs. These insights underscore the critical role of ES

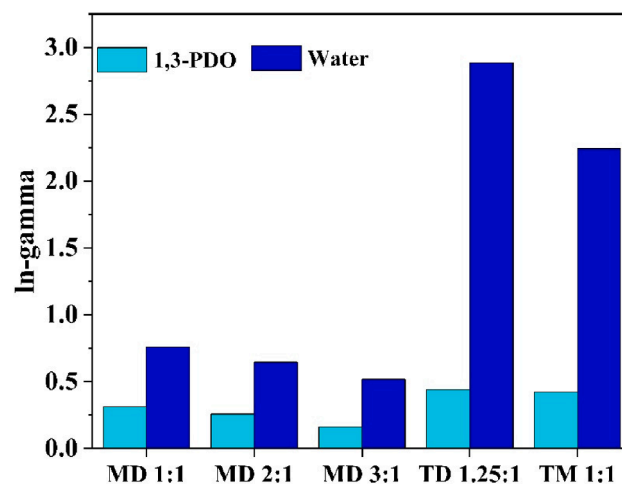


Fig. 9. Logarithmic infinite dilution activity coefficients (ln-gamma) of 1,3-PDO and Water in ESs.

**Table 6**

Solvents Studied in literature for the extraction of 1,3-PDO from aqueous solution.

| Solvents                                                                                               | $\beta_{1,3\text{-PDO}}$ | $S_{1,3\text{-PDO}}$ | Ref. | Solvents                                                                           | $\beta_{1,3\text{-PDO}}$ | $S_{1,3\text{-PDO}}$ | Ref. |
|--------------------------------------------------------------------------------------------------------|--------------------------|----------------------|------|------------------------------------------------------------------------------------|--------------------------|----------------------|------|
| [Bmim] <sup>+</sup> [NTf2] <sup>−</sup>                                                                | 0.015                    | 1.08                 | [11] | [Pl <sub>4,1</sub> ] <sup>+</sup> [CF <sub>3</sub> SO <sub>3</sub> ] <sup>−</sup>  | 7.13                     | —                    | [8]  |
| [PNMIM] <sup>+</sup> [NTf2] <sup>−</sup>                                                               | 0.024                    | 0.62                 | [11] | [Im <sub>2,1</sub> ] <sup>+</sup> [CF <sub>3</sub> SO <sub>3</sub> ] <sup>−</sup>  | 12.31                    | —                    | [8]  |
| [BOOMMIM] <sup>+</sup> [NTf2] <sup>−</sup>                                                             | 0.019                    | 0.89                 | [11] | [Im <sub>20,1</sub> ] <sup>+</sup> [CF <sub>3</sub> SO <sub>3</sub> ] <sup>−</sup> | 12.00                    | —                    | [8]  |
| [BPy] <sup>+</sup> [NTf2] <sup>−</sup>                                                                 | 0.010                    | 0.78                 | [11] | Ethyl acetate and ethanol cosolvent (9:1)                                          | 0.20                     | 1.17                 | [7]  |
| [P <sub>66614</sub> ] <sup>+</sup> [1-octane-sulfonate (C <sub>8</sub> SO <sub>3</sub> )] <sup>−</sup> | 0.219                    | 2.65                 | [11] | Ethanol and K <sub>2</sub> HPO <sub>4</sub>                                        | —                        | 4.870                | [10] |
| [P <sub>66614</sub> ] <sup>+</sup> [diisooctylphosphinate (DIOP)] <sup>−</sup>                         | 0.333                    | 2.05                 | [11] | Ethanol and saturated ammonium sulfate – 46 % (v/v)                                | 4.77                     | —                    | [78] |
| [P <sub>66614</sub> ] <sup>+</sup> [salicylate (SAL)] <sup>−</sup>                                     | 0.144                    | 2.06                 | [11] | NaH <sub>2</sub> PO <sub>4</sub> (5 wt%) and n-butyl acetate (30 wt%)              | 9.40                     | —                    | [79] |
| [P <sub>66614</sub> ] <sup>+</sup> [DCA] <sup>−</sup>                                                  | 0.095                    | 3.15                 | [11] | Pentanol                                                                           | 0.25                     | 2.10                 | [80] |
| [P <sub>66614</sub> ] <sup>+</sup> [SCN] <sup>−</sup>                                                  | 0.129                    | 3.06                 | [11] | Hexanol                                                                            | 0.16                     | 1.94                 | [80] |
| [P <sub>66614</sub> ] <sup>+</sup> Cl <sup>−</sup>                                                     | 0.334                    | 2.11                 | [11] | Heptanol                                                                           | 0.11                     | 1.77                 | [80] |
| [PNMIM] <sup>+</sup> [NTf2] <sup>−</sup>                                                               | 0.024                    | 0.62                 | [11] | Octanol                                                                            | 0.09                     | 1.68                 | [80] |
| Imidazolium based ILs with K <sub>2</sub> HPO <sub>4</sub> and KH <sub>2</sub> PO <sub>4</sub>         |                          |                      | [8]  | Nonanol                                                                            | 0.07                     | 1.97                 | [80] |
| [Im <sub>4,1</sub> ] <sup>+</sup> [CF <sub>3</sub> SO <sub>3</sub> ] <sup>−</sup>                      | 10.24                    | —                    | [8]  | Hexanal                                                                            | 0.28                     | 20.18                | [80] |
| [Im <sub>4,1</sub> ] <sup>+</sup> [N(CN) <sub>2</sub> ] <sup>−</sup>                                   | 12.61                    | —                    | [8]  | Nonanal                                                                            | 0.11                     | 7.85                 | [80] |
| [Im <sub>4,1</sub> ] <sup>+</sup> [SCN] <sup>−</sup>                                                   | 12.73                    | —                    | [8]  | Decanal                                                                            | 0.09                     | 31.11                | [80] |
| [Im <sub>4,1</sub> ] <sup>+</sup> [CH <sub>3</sub> SO <sub>4</sub> ] <sup>−</sup>                      | 26.01                    | —                    | [8]  |                                                                                    |                          |                      |      |
| [Mo <sub>4,1</sub> ] <sup>+</sup> [CF <sub>3</sub> SO <sub>3</sub> ] <sup>−</sup>                      | 4.54                     | —                    | [8]  |                                                                                    |                          |                      |      |

composition in influencing thermodynamic properties relevant to extraction processes, guiding the optimization of solvent selection and design for enhanced industrial performance.

#### 4.9. Contextualizing findings within the literature

When comparing the present work's experimental data to the available literature values, it is evident that the  $\beta_{1,3\text{-PDO}}$  values reported in the present work, ranging from 20 to 233, surpass the best-reported value in the literature (Table 6). In another study investigating the ability of different imidazolium-based ionic liquids (ILs) to form aqueous two-phase systems (ATPSs) with K<sub>2</sub>HPO<sub>4</sub> and KH<sub>2</sub>PO<sub>4</sub>, the distribution coefficient  $\beta_{1,3\text{-PDO}}$  was measured. The IL [Im<sub>4,1</sub>]<sup>+</sup>[CH<sub>3</sub>SO<sub>4</sub>]<sup>−</sup> demonstrated the highest distribution coefficient with a value of 26.01. Additionally, an aqueous two-phase system comprising NaH<sub>2</sub>PO<sub>4</sub> (5 wt%) and n-butyl acetate (30 wt%) demonstrates a  $\beta_{1,3\text{-PDO}}$  of 9.40. In a study by Malinowski et al. (1999), several conventional solvents were evaluated for separating 1,3-PDO from the aqueous phase, with hexanal showing a  $\beta_{1,3\text{-PDO}}$  of 0.28. This comparison highlights the superior separation efficiency of the eutectic solvents studied in the present work. Furthermore, the  $S_{1,3\text{-PDO}}$  values of the present work, which range from 482 to 17991, far exceed those reported in the literature. Moreover, The MD 3:1 system exhibits the highest  $\beta_{1,3\text{-PDO}}$  value of 233.89 and an  $S$  value of 17991, positioning it as the most effective solvent among all the investigated solvents.

This study primarily focuses on evaluating phase equilibria and, solute-solvent interaction, with particular emphasis on the performance of ESs in extracting 1,3-PDO from aqueous solution. Additionally, assessing the selectivity of ESs in extracting 1,3-PDO relative to other components in the fermentation broth is crucial. Future research is expected to involve scaled-up studies to assess the techno-economic feasibility of the process, as well as Life Cycle Assessment (LCA) to evaluate the environmental impact of ES-based processes.

## 5. Conclusions

Hydrophobic eutectic solvents were successfully synthesized in the desired ratios, confirmed to be liquid at room temperature, and characterized using FTIR, DSC, and NMR spectroscopy to verify their compositions. The present research on the phase equilibria of ES/1,3-PDO/water systems has provided significant insights into the extraction efficiency of selected ESs for liquid-liquid extraction (LLE) from aqueous systems. The study was meticulously validated through molecular dynamics (MD) simulations, ensuring robustness in our findings. Among the various ES compositions analyzed, DL-menthol dodecanoic acid-based DES at a 3:1 ratio exhibited the least diffusion of water into the

ES phase, followed by MD 2:1, MD 1:1, and TM 1:1. Conversely, the TD 1.25:1 ES displayed the highest diffusion of water into the ES phase. The density profile and volume slice plots consistently demonstrated a clear interface between the water and ES phases, affirming the phase separation. Notably, the trends observed in our experimental results were in close agreement with the computational simulations, underscoring the reliability and predictive capability of the simulation. Among the solvents tested, MD 3:1 stands out as the most effective solvent for extracting 1,3-PDO from aqueous solutions, offering the highest  $\beta$  and  $S$  of 233.89 and 17991, respectively. MD 2:1 also shows strong performance, followed by MP 12:1, TM 1:1, MD 1:1, and TD 1.25:1 exhibit varying degrees of moderate to lower extraction efficiency and selectivity. These findings provide essential guidance for selecting optimal solvents to enhance the extraction process of 1,3-PDO in industrial applications. Furthermore, MD simulations and COSMO-SAC analysis corroborate the experimental results, following the same trend. The RDF, SDF, hydrogen bonding analysis and self-diffusion coefficient justify these results. MD 3:1, characterized by a higher number of –OH groups, exhibits enhanced intermolecular hydrogen bonding compared to other studied ESs. This property influences the interaction dynamics between the ES and 1,3-PDO, facilitating stronger binding and potential entrapment of 1,3-PDO molecules within the MD 3:1 DES phase. The COSMO-SAC model shows that ESs compositions greatly influence the solvation of 1,3-PDO and water. Lower infinite activity coefficient values for 1,3-PDO, especially in MD 3:1 DES, indicate strong solute-solvent interactions, aiding efficient extraction. Higher infinite activity coefficient values for water suggest weaker interactions in ES, emphasizing ES's specificity for extracting organic compounds like 1,3-PDO from water. Moreover, the present work did not detect any solvent loss, as evidenced by the absence of prominent ES peaks in the raffinate phase in <sup>1</sup>H NMR spectra. Additionally, MD simulations indicated no trace of ES entering the water phase, further affirming the effective separation and containment of ES within the system. Therefore, cost-effective hydrophobic ESs present a promising alternative to conventional solvents and ILs. This work further shows the potential of tailored ESs in optimizing extraction processes, paving the way for more efficient extraction of 1,3-PDO.

#### CRedit authorship contribution statement

**Raj Akshat:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Anand Bharti:** Writing – review & editing, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Padmini Padmanabhan:** Writing – review &

editing, Supervision, Resources, Project administration, Methodology, Conceptualization.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.molliq.2025.127026>.

### Data availability

Data will be made available on request.

### References

- [1] Liu J zhong, Xu W, Chistoserdiv A, Bajpai RK. Glycerol Dehydratases: Biochemical Structures, Catalytic Mechanisms, and Industrial Applications in 1,3-Propanediol Production by Naturally Occurring and Genetically Engineered Bacterial Strains. *Appl Biochem Biotechnol* . 2016 Jul 1;179(6):1073–100. Available from: <https://link.springer.com/article/10.1007/s12010-016-2051-6>.
- [2] P. Anand, R.K. Saxena, R.G. Marwah, A novel downstream process for 1,3-propanediol from glycerol-based fermentation, *Appl Microbiol Biotechnol*. 90 (4) (2011 May) 1267–1276.
- [3] 1,3 Propanediol Market Size, Share & Growth Report, 2030 . Available from: <https://www.grandviewresearch.com/industry-analysis/1-3-propanediol-pdo-market>.
- [4] J.V. Kurian, A new polymer platform for the future - Sorona® from corn derived 1,3-propanediol, Available from, *J Polym Environ* . 13 (2) (2005 Apr) 159–167, <https://link.springer.com/article/10.1007/s10924-005-2947-7>.
- [5] G.A. Kraus, Synthetic Methods for the Preparation of 1,3-Propanediol, Available from, *Clean (weinh)* . 36 (8) (2008 Aug 1) 648–651.
- [6] R.K. Saxena, P. Anand, S. Saran, J. Isar, Microbial production of 1,3-propanediol: Recent developments and emerging opportunities, *Biotechnol Adv*. 27 (6) (2009 Aug) 895–913.
- [7] X.L. Wang, J.J. Zhou, Y.Q. Sun, Z.L. Xiu, Bioconversion of raw glycerol from waste cooking-oil-based biodiesel production to 1,3-propanediol and lactate by a microbial consortium. *Front Bioeng, Biotechnol* . (2019). Feb 18;7(FEB):426245. Available from: [www.frontiersin.org](http://www.frontiersin.org).
- [8] T. Boonsongsawat, A. Shotipruk, V. Tantayakom, P. Prasitchoke, C. Chandavasu, P. Boonnoun, et al., Solvent extraction of biologically derived 1,3-propanediol with ethyl acetate and ethanol cosolvent, *Sep Sci Technol*. 45 (4) (2010 Jan) 541–547.
- [9] A. Müller, A. Górak, Extraction of 1,3-propanediol from aqueous solutions using different ionic liquid-based aqueous two-phase systems, *Sep. Purif. Technol*. (2012) 130–136.
- [10] W.B. Sui, Y.Q. Sun, X.L. Wang, Z.L. Xiu, Synergistic Extraction of 1,3-Propanediol from Fermentation Broths Using Multialcohol Extractants, *ACS Sustain Chem Eng*. 10 (36) (2022 Sep 12) 11891–11901.
- [11] V. Liberato, F.S. Buarque, T.F. Ferreira, B.D. Ribeiro, M.A.Z. Coelho, Salting-out as a Fast and Simple Method for First-Step Separation of Biobased 1,3-Propanediol, Available from, *J Chem Eng Data* . (2023).
- [12] X.H. Liu, M. Rebros, I. Dolejš, A.C. Marr, Designing Ionic Liquids for the Extraction of Alcohols from Fermentation Broth: Phosphonium Alkanesulfonates, Solvents for Diol Extraction, *ACS Sustain Chem Eng*. 5 (9) (2017 Sep 5) 8260–8268.
- [13] X. Zhao, H. Wu, M. Duan, X. Hao, Q. Yang, Q. Zhang, et al., Liquid-liquid extraction of lithium from aqueous solution using novel ionic liquid extractants via COSMO-RS and experiments, *Fluid Phase Equilib*. 15 (459) (2018 Mar) 129–137.
- [14] X. Wu, Z. Lei, Q. Li, J. Zhu, B. Chen, Liquid–Liquid Extraction of Low-Concentration Aniline from Aqueous Solutions with Salts, Available from, *Ind Eng Chem Res* . 49 (6) (2010 Mar 17) 2581–2588.
- [15] R. Akshat, A. Bharti, P. Padmanabhan, Atomistic molecular dynamics simulation and COSMO-SAC approach for enhanced 1,3-propanediol extraction with imidazolium-based ionic liquids, *J Mol Model*. 30 (6) (2024 Jun 1) 164.
- [16] C.M.S.S. Neves, J.F.O. Granjo, M.G. Freire, A. Robertson, N.M.C. Oliveira, J.A. P. Coutinho, Separation of ethanol–water mixtures by liquid–liquid extraction using phosphonium-based ionic liquids, *Green Chem*. 13 (6) (2011 Jun 7) 1517–1526.
- [17] W.Y. Lee, K.S. Kim, J.K. You, Y.K. Hong, Effect of Cations in Ionic Liquids on the Extraction Characteristics of 1,3-Propanediol by Ionic Liquid-based Aqueous Biphasic Systems, *ACS Sustain Chem Eng*. 4 (2) (2016 Feb 1) 572–576.
- [18] E.L. Smith, A.P. Abbott, K.S. Ryder, Deep Eutectic Solvents (DESs) and Their Applications, Available from, *Chem Rev* . 114 (21) (2014 Nov 12) 11060–11082.
- [19] A. Paiva, R. Craveiro, I. Aroso, M. Martins, R.L. Reis, A.R.C. Duarte, Natural deep eutectic solvents - Solvents for the 21st century, Available from, *ACS Sustain Chem Eng* . 2 (5) (2014 May 5) 1063–1071.
- [20] Y. Dai, J. van Spronsen, G.J. Witkamp, R. Verpoorte, Y.H. Choi, Natural deep eutectic solvents as new potential media for green technology, *Anal Chim Acta*. 5 (766) (2013 Mar) 61–68.
- [21] Q. Zhang, V.K. De Oliveira, S. Royer, F. Jérôme, Deep eutectic solvents: syntheses, properties and applications, Available from, *Chem Soc Rev* . 41 (21) (2012 Oct 8) 7108–7146.
- [22] C. Florindo, L.C. Branco, I.M. Marrucho, Quest for Green-Solvent Design: From Hydrophilic to Hydrophobic (Deep) Eutectic Solvents, Available from, *ChemSusChem* 12 (8) (2019 Apr 23) 1549–1559.
- [23] P.K. Naik, S. Paul, T. Banerjee, Liquid Liquid Equilibria measurements for the extraction of poly aromatic nitrogen hydrocarbons with a low cost Deep Eutectic Solvent: Experimental and theoretical insights, *J Mol Liq*. 1 (243) (2017 Oct) 542–552.
- [24] K. Yin, Y. Shen, Y. Jiao, M. Zhou, X. Wang, Z. Zhu, et al., Mechanism analysis of liquid-liquid extraction separation for dimethyl carbonate and methanol by choline-based DESs as extractants, *Fluid Phase Equilib*. 1 (566) (2023 Mar) 113701.
- [25] C. Florindo, L.C. Branco, I.M. Marrucho, Development of hydrophobic deep eutectic solvents for extraction of pesticides from aqueous environments, *Fluid Phase Equilib*. 25 (448) (2017 Sep) 135–142.
- [26] E. Riveiro, B. González, Á. Domínguez, Extraction of adipic, levulinic and succinic acids from water using TOPO-based deep eutectic solvents, *Sep Purif Technol*. 15 (241) (2020 Jun) 116692.
- [27] R. Verma, P.K. Naik, I. Diaz, T. Banerjee, Separation of low molecular weight alcohols from water with deep eutectic solvents: Liquid-liquid equilibria and process simulations, *Fluid Phase Equilib*. 1 (533) (2021 Apr) 112949.
- [28] N. Schaeffer, M.A.R. Martins, C.M.S.S. Neves, S.P. Pinho, J.A.P. Coutinho, Sustainable hydrophobic terpene-based eutectic solvents for the extraction and separation of metals, Available from, *Chem. Commun*. 54 (58) (2018 Jul 17) 8104–8107.
- [29] D. Rodríguez-Llorente, A. Bengoa, G. Pascual-Muñoz, P. Navarro, V.I. Águeda, J. A. Delgado, et al., Sustainable Recovery of Volatile Fatty Acids from Aqueous Solutions Using Terpenoids and Eutectic Solvents, Available from, *ACS Sustain Chem Eng* . 7 (19) (2019 Oct 7) 16786–16794.
- [30] D.J.G.P. Van Osch, L.F. Zubeir, A. Van Den Bruinhorst, M.A.A. Rocha, M.C. Kroon, Hydrophobic deep eutectic solvents as water-immiscible extractants, Available from, *Green Chem*. 17 (9) (2015 Sep 1) 4518–4521.
- [31] Verma R, Banerjee T, Verma R, Banerjee T. Palmitic-Acid-Based Hydrophobic Deep Eutectic Solvents for the Extraction of Lower Alcohols from Aqueous Media: Liquid–Liquid Equilibria Measurements, Validation and Process Economics. *Global Challenges* . 2019 Nov 1 ;3(11):1900024. Available from: <https://onlinelibrary.wiley.com/doi/full/10.1002/gch2.201900024>.
- [32] R. Verma, M. Mohan, V.V. Goud, T. Banerjee, Operational Strategies and Comprehensive Evaluation of Menthol Based Deep Eutectic Solvent for the Extraction of Lower Alcohols from Aqueous Media, *ACS Sustain Chem Eng*. 6 (12) (2018 Dec 3) 16920–16932.
- [33] Lalikoglu M. Separation of butyric acid from aqueous media using menthol-based hydrophobic deep eutectic solvent and modeling by response surface methodology. *Biomass Convers Biorefin* . 2022;12:3. Available from: <https://doi.org/10.1007/s13399-021-01711-7>.
- [34] J. Plotka-Wasyłka, M. de la Guardia, V. Andrich, M. Vilková, Deep eutectic solvents vs ionic liquids: Similarities and differences, *Microchem. J*. 1 (159) (2020 Dec) 105539.
- [35] Y. Wu, D. Meng, D. Yao, X. Liu, Y. Xu, Z. Zhu, et al., Mechanism Analysis, Economic Optimization, and Environmental Assessment of Hybrid Extractive Distillation-Pervaporation Processes for Dehydration of n-Propanol, Available from, *ACS Sustain Chem Eng* . 8 (11) (2020 Mar 23) 4561–4571.
- [36] J.M. Otero-Mato, H. Montes-Campos, V. Gómez-González, M. Montoto, O. Cabeza, S. Kondrat, et al., Structure, dynamics and conductivities of ionic liquid-alcohol mixtures, *J Mol Liq*. 1 (2022 Jun) 355.
- [37] Argüello-Luengo J, González-Tudela A, Shi T, Zoller P, Cirac JI. Analogue quantum chemistry simulation. *Nature* 2019 574:7777 . 2019 Oct 9 ;574(7777):215–8. Available from: <https://www.nature.com/articles/s41586-019-1614-4>.
- [38] Y. Cao, J. Romero, J.P. Olson, M. Degroote, P.D. Johnson, M. Kieferová, et al., Quantum Chemistry in the Age of Quantum Computing, Available from, *Chem Rev* . 119 (19) (2019 Oct 9) 10856–10915.
- [39] R. Li, S. Chu, C. Cui, K. Jia, J. Li, Study on separation of methanol and dimethyl carbonate azeotropic system with deep eutectic solvents, *J Mol Liq*. 1 (399) (2024 Apr) 124301.
- [40] Li J, Wang J, Wu M, Cheng H, Chen L, Qi Z. Deep Deterpenation of Citrus Essential Oils Intensified by In Situ Formation of a Deep Eutectic Solvent in Associative



- Extraction. 2020 May 13 ; Available from: /articles/journal\_contribution/Deep\_Deterpenation\_of\_Citrus\_Essential\_Oils\_Intensified\_by\_In\_Situ\_Formation\_of\_a\_Deep\_Eutectic\_Solvent\_in\_Associative\_Extraction/12231125/1.
- [41] P.K. Naik, D. Kundu, P. Bairagya, T. Banerjee, Phase behavior of water-menthol based deep eutectic solvent-dodecane system, *Chem. Thermodyn. Therm. Anal.* 1 (2021 Sep) 3–4.
  - [42] U.L. Abbas, Q. Qiao, M.T. Nguyen, J. Shi, Q. Shao, Structure and hydrogen bonds of hydrophobic deep eutectic solvent-aqueous liquid-liquid interfaces, *AIChE J* (2021). Dec 1;67(12).
  - [43] S. Tanaka, Theory of power-compensated DSC, *Thermochim. Acta.* (1992). Nov 30; 210(C):67–76.
  - [44] Jorgensen WL, Maxwell DS, Tirado-Rives J. Development and Testing of the OPLS All-Atom Force Field on Conformational Energetics and Properties of Organic Liquids . 1996. Available from: <https://pubs.acs.org/sharingguidelines>.
  - [45] M. Udier-Blagović, P. Morales De Tirado, S.A. Pearman, W.L. Jorgensen, Accuracy of free energies of hydration using CM1 and CM3 atomic charges, Available from: *J Comput Chem* . 25 (11) (2004 Aug 1) 1322–1332.
  - [46] L.S. Dodda, J.Z. Vilseck, J. Tirado-Rives, W.L. Jorgensen, 1.14\*CM1A-LBCC: Localized Bond-Charge Corrected CM1A Charges for Condensed-Phase Simulations, *J. Phys. Chem. B* 121 (15) (2017 Apr 20) 3864–3870.
  - [47] L.S. Dodda, I.C. De Vaca, J. Tirado-Rives, W.L. Jorgensen, LigParGen web server: An automatic OPLS-AA parameter generator for organic ligands, *Nucleic Acids Res.* 45 (W1) (2017 Jul 3) W331–W336.
  - [48] D. Vassetz, M. Pagliai, P. Procacci, Assessment of GAFF2 and OPLS-AA General Force Fields in Combination with the Water Models TIP3P, SPCE, and OPC3 for the Solvation Free Energy of Druglike Organic Molecules, Available from, *J Chem Theory Comput* . 15 (3) (2019 Feb 14) 1983–1995.
  - [49] Izadi S, Onufriev A V. Accuracy limit of rigid 3-point water models. *J Chem Phys* . 2016 Aug 21 ;145(7). Available from: <https://pubmed.ncbi.nlm.nih.gov/27544113/>.
  - [50] S.P. Kadaoluwa Pathirannahalage, N. Meftahi, A. Elbourne, A.C.G. Weiss, C. F. McConville, A. Padua, et al., Systematic Comparison of the Structural and Dynamic Properties of Commonly Used Water Models for Molecular Dynamics Simulations, *J Chem Inf Model.* 61 (9) (2021 Sep 27) 4521–4536.
  - [51] L. Martinez, R. Andrade, E.G. Birgin, J.M. Martinez, PACKMOL: A package for building initial configurations for molecular dynamics simulations, *J Comput Chem.* 30 (13) (2009) 2157–2164.
  - [52] K. Li, X. Lin, J. Greenberg, Software citation, reuse and metadata considerations: An exploratory study examining LAMMPS, *Proc. Assoc. Inf. Sci. Technol.* 53 (1) (2016) 1–10.
  - [53] Luty BA, Davis ME, Tironi IG, Van Gunsteren WF. A Comparison of Particle-Particle, Particle-Mesh and Ewald Methods for Calculating Electrostatic Interactions in Periodic Molecular Systems. <https://doi.org/10.1080/08927029408022004> . 2006 ;14(1):11–20. Available from: <https://www.tandfonline.com/doi/abs/10.1080/08927029408022004>.
  - [54] Nosé S, Klein ML. A study of solid and liquid carbon tetrafluoride using the constant pressure molecular dynamics technique. *J Chem Phys* . 1983 Jun 1 ;78 (11):6928–39. Available from: /aip/jcp/article/78/11/6928/439000/A-study-of-solid-and-liquid-carbon-tetrafluoride.
  - [55] W.G. Hoover, Canonical dynamics: Equilibrium phase-space distributions, *Mar* 1; 31(3):1695. Available from: *Phys Rev A (coll Park)* (1985).
  - [56] Parrinello M, Rahman A. Polymorphic transitions in single crystals: A new molecular dynamics method. *J Appl Phys* . 1981 Dec 1;52(12):7182–90. Available from: /aip/jap/article/52/12/7182/9558/Polymorphic-transitions-in-single-crystals-A-new.
  - [57] Brehm M, Thomas M, Gehrke S, Kirchner B. TRAVIS—A free analyzer for trajectories from molecular simulation. *Journal of Chemical Physics* . 2020 Sep;152 (16):164105. Available from: /aip/jcp/article/152/16/164105/198859/TRAVIS-A-free-analyzer-for-trajectories-from.
  - [58] W. Humphrey, A. Dalke, K. Schulten, VMD – Visual Molecular Dynamics, *J Mol Graph.* 14 (1996) 33–38.
  - [59] G.E. Logotheti, J. Ramos, I.G. Economou, Molecular modeling of imidazolium-based [Tf2N-] ionic liquids: Microscopic structure, thermodynamic and dynamic properties, and segmental dynamics, Available from, *J. Phys. Chem. B* 113 (20) (2009 Aug) 7211–7224.
  - [60] A. Klamt, Conductor-like screening model for real solvents: A new approach to the quantitative calculation of solvation phenomena, Available from, *J. Phys. Chem.* 99 (7) (1995) 2224–2235.
  - [61] A. Klamt, F. Eckert, COSMO-RS: a novel and efficient method for the a priori prediction of thermophysical data of liquids, *Fluid Phase Equilib.* 172 (1) (2000 Jul 28) 43–72.
  - [62] A. Klamt, S.T. Lin, S.I. Sandler, Comments on “A priori phase equilibrium prediction from a segment contribution solvation model” (Multiple letters), Available from, *Ind Eng Chem Res* . 41 (9) (2002 May 1) 2330–2334.
  - [63] M.D. Hanwell, D.E. Curtis, D.C. Lonie, T. Vandermeersch, E. Zurek, G. R. Hutchison, Avogadro: an advanced semantic chemical editor, visualization, and analysis platform, Available from: *J Cheminform* . 4 (1) (2012 Aug).
  - [64] M.J. Frisch, G.W. Trucks, H.B. Schlegel, G.E. Scuseria, M.A. Robb, J.R. Cheeseman, et al., Gaussian16 Revision C.01. (2016).
  - [65] C. Lee, W. Yang, R.G. Parr, Development of the Colle-Salvetti correlation-energy formula into a functional of the electron density, *Jan* 15;37(2):785. Available from: *PhysRevB*. (1988).
  - [66] A.D. Becke, Density-functional thermochemistry. III. The role of exact exchange, *J. Chem. Phys.* 98 (7) (1993) 5648–5652.
  - [67] I.H. Bell, E. Mickleit, C.M. Hsieh, S.T. Lin, J. Vrabec, C. Breitkopf, et al., A Benchmark Open-Source Implementation of COSMO-SAC, Available from, *J Chem Theory Comput* . 16 (4) (2020 Apr 14) 2635–2646.
  - [68] C.M. Hsieh, S.I. Sandler, S.T. Lin, Improvements of COSMO-SAC for vapor-liquid and liquid-liquid equilibrium predictions, *Fluid Phase Equilib.* 297 (1) (2010 Oct 15) 90–97.
  - [69] B.D. Ribeiro, C. Florindo, L.C. Iff, M.A.Z. Coelho, I.M. Marrucho, Menthol-based eutectic mixtures: Hydrophobic low viscosity solvents, Available from, *ACS Sustain Chem Eng* . 3 (10) (2015 Sep 10) 2469–2477.
  - [70] D.O. Adeoye, Z.S. Gano, O.U. Ahmed, S.M. Shuwa, A.Y. Atta, S.A. Iwarere, et al., Synthesis and Characterisation of Menthol-Based Hydrophobic Deep Eutectic Solvents, In MDPI AG (2024) 98.
  - [71] D. Ge, Y. Zhang, Y. Dai, S. Yang, Air-assisted dispersive liquid-liquid microextraction based on a new hydrophobic deep eutectic solvent for the preconcentration of benzophenone-type UV filters from aqueous samples, Available from, *J Sep Sci* . 41 (7) (2018 Apr 1) 1635–1643.
  - [72] Bergua F, Castro M, Lafuente C, Artal M. Thymol+I-menthol eutectic mixtures: Thermophysical properties and possible applications as decontaminants. Available from: <http://creativecommons.org/licenses/by/4.0/>.
  - [73] D. Xu, J. Chow, C.C. Weber, M.A. Packer, S. Baroutian, K. Shahbaz, Evaluation of deep eutectic solvents for the extraction of fucoxanthin from the alga *Tisochrysis lutea* – COSMO-RS screening and experimental validation, *J Environ Chem Eng.* 10 (5) (2022 Oct 1) 108370.
  - [74] R. Zhang, X. Yue, B. Li, J. Yang, Z. Wu, J. Zhang, Dynamic viscosity, density and surface tension of 1,3-propanediol (1) + 1,2-propanediamine (2) binary system at T = (293.15 to 318.15) K and atmosphere pressure, *J Mol Liq.* (2020). Feb 1;299.
  - [75] Luzar A, Chandler D. Hydrogen-bond kinetics in liquid water. *Nature* 1996 379: 6560 . 1996 Jan 4;379(6560):55–7. Available from: <https://www.nature.com/articles/379055a0>.
  - [76] Luzar A. Resolving the hydrogen bond dynamics conundrum. *J Chem Phys* . 2000 Dec 15 ;113(23):10663–75. Available from: /aip/jcp/article/113/23/10663/183853/Resolving-the-hydrogen-bond-dynamics-conundrum.
  - [77] W. Zhang, Z. Chen, Y. Shen, G. Li, Y. Dai, J. Qi, et al., Molecular Mechanism and Extraction Performance Evaluation for Separation of Methanol and n-Hexane via Ionic Liquids as Extractant, *ACS Sustain Chem Eng.* 8 (23) (2020 Jun 15) 8700–8712.
  - [78] Z. Li, B. Jiang, D. Zhang, Z. Xiu, Aqueous two-phase extraction of 1,3-propanediol from glycerol-based fermentation broths, *Sep Purif Technol.* 66 (3) (2009 May 7) 472–478.
  - [79] Z. Li, L. Yan, J. Zhou, X. Wang, Y. Sun, Z.L. Xiu, Two-step salting-out extraction of 1,3-propanediol, butyric acid and acetic acid from fermentation broths, *Sep Purif Technol.* 31 (209) (2019 Jan) 246–253.
  - [80] J.J. Malinowski, Evaluation of liquid extraction potentials for downstream separation of 1,3-propanediol, *Biotechnol. Tech.* Vol. 13 (1999).